

# Solution, Exam: FM1015 Modelling of Dynamic Systems

Date: Friday, December 1, 2023

*Form of exam:* This is a *Digital Exam* of type *Multiple Choice*. Each exam *question* (denoted “problems” in the text below) is followed by *4 possible answers* (“choice” in the tables related to questions). *Some* details about your answers and how they will be assessed:

- A single correct choice gives full score on a problem; any other choice (an incorrect choice, no choice, several choices) gives zero score on the particular problem.
- All problems are given the same weight.
- Most problems are posed so that it is necessary to do background calculations to choose the correct answer (A–D).
- The background calculations shall not be submitted — the only submission in the Digital Exam are the choices (A–D) in each problem.
- You need to insert the answers in the *attached Excel* sheet, and upload this *Excel* sheet which includes your answers, in the exam system.
- The file you **upload must be in Excel file format**, i.e., \*.xlsx format. (*Reason:* the files will be automatically parsed for grading.)
- A total score of 0-39% gives grade F (fail), score 40-49% gives grade E, score 50-59% gives grade D, score 60-79% gives grade C, score 80-89% gives grade B, and score 90-100% gives grade A.

*Aids:* No aids allowed, except the necessary use of PC for using WISEflow, and the use of Excel for filling in answers.

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This exam contains questions (denoted *problems*) in the *Problems* part. The Problems part consists of *20 problems*, each with 4 alternative choices (A–D).

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## Comments to exam

1. When the exam was designed, it was the intention that you, the candidates, should be allowed to use MATLAB or Python or similar computational tools on the exam. However, the exam administration claimed that I could not do that, because they said the official web-page for the course say “no aids allowed”. So I followed their decision. In hindsight, I checked the web page, and it actually says “no aids allowed normally”.
2. Because some tasks perhaps would be easier to solve with access to, say, Python, I have looked over the tasks again to check whether there are uncertainties, etc. in the problems, and whether the lack of access to use of, say, Python, has lead to poorer score. Here is my assessment on the effect of no access to Python, etc.:
  - (a) Problems 1, 2: *no effect*.
  - (b) Problem 3: use of Python, or similar, would certain be beneficial. It is always possible to make an intelligent guess, but use of a computer tool would have made it possible to know the answer for certain. *Perhaps a significant effect*.
  - (c) Problem 4: use of Python, etc., would be beneficial. At the same time, those who worked with the project and read the solution should know that the effect is small (i.e., in the range of tens of mV), and should be able to figure out the sign. *So a minor effect*.
  - (d) Problems 5, 6, 7, 8: *no effect*.
  - (e) Problem 9: *no effect*. One candidate claimed that I should have defined the `max()` and `min()` functions, and suggested that these could be interpreted in two different ways. I disagree: in my long life, I have only seen a single definition of these functions. The alternative definition suggested by the candidate is never used; perhaps he/she confuses these function with the `minmax()` function, which is something totally different.
  - (f) Problem 10: *no effect*. One candidate claimed there is no correct answer to this problem. The confusion is probably due to a change in definition of  $\nu_{\text{H}_2}$  compared to the project. NOTE: there is a typo in the alternatives: alternatives A and D were the same, and both were wrong. This, in fact, makes it *easier* to solve this problem, so is an advantage to you, the candidates. In this solution, I have marked in red and fixed the alternative that should have been different.
  - (g) Problem 11–16: *no effect*.
  - (h) Problem 17: *a minor effect*. From the theoretical expression given (Problem 17 + Problem 2), it should be straightforward to find that  $\eta = \Delta_r \tilde{G}^\circ / \Delta_r \tilde{H}^\circ$ . With the numerator provided as  $\Delta_r \tilde{G}^\circ = -228.572 \text{ kJ/mol}$  and  $\Delta_r \tilde{H}^\circ = -241.785 \text{ kJ/mol}$ ,  $\eta \approx 229/242 \approx 236/250 \approx 945/1000$  which is very close to 0.95. So this is doable without a computer.
  - (i) Problem 18: *a minor effect*. From Figure 7, there can be little doubt that at  $i'' = 0.5 \text{ A/cm}^2$ , voltage  $u$  is approximately 50% of the value of the Nernst standard state voltage (blue dashed line).
  - (j) Problem 19–20: *no effect*.

3. In summary: lack of access to Python, MATLAB, Julia, or similar would have perhaps a *significant effect* in Problem 3, and a *minor effect* in Problems 4, 17, 18. On the other hand, Problem 10 had a typo which made it simpler to solve.
4. It turns out that no-one had Problem 3 correct. In summary, I have chosen to give everyone a full score on Problem 3.

With these conditions in mind, the grade distribution is as follows:

1×11 DataFrame

| Row | A     | B     | C     | D     | E     | F     | abs   |
|-----|-------|-------|-------|-------|-------|-------|-------|
|     | Int64 | Int64 | Int64 | Int64 | Int64 | Int64 | Int64 |
| 1   | 5     | 7     | 26    | 12    | 5     | 3     | 0     |

## Problems

Hydrogen (H<sub>2</sub>) and oxygen (O<sub>2</sub>), when mixed together, form an explosive mixture<sup>1</sup> which reacts quickly and violently. If mixed with other, inert gases such as nitrogen (e.g., in air), the mixture is less explosive, and may become harmless when the partial pressure of hydrogen is below a certain threshold value.

It is of interest to try to control this violent reaction so that we can utilize the energy in the mixture. This can be done by putting the reacting gases (H<sub>2</sub>, O<sub>2</sub>) in different volumes, separated by a *membrane*. Such a membrane may be physically damaged if the *total pressure difference across* the membrane is too large (for the fuel cell discussed below, the pressure difference should be less than ca. 0.2 bar). Also, if the *partial pressure difference* for one gas is large across the membrane, this particular gas may “leak” (permeate) through the membrane.

A fuel cell can be used to take out the energy of hydrogen and oxygen: a certain type of membrane that under normal conditions blocks permeation of any species except *proton* (H<sup>+</sup>) can be used (PEM = Proton Exchange Membrane). Then some half-reactions occur on each side of the membrane, accelerated by a catalyst. The fuel cell includes a system for feeding of reactants (H<sub>2</sub>, O<sub>2</sub>) to each side of the membrane, and for removing the product (H<sub>2</sub>O). Note that normal PEM systems require the membrane to be humid (= contain some water) in order to work. Because of this, some water is normally added to both sides of the membrane.

### Problem 1. Reactions in Fuel Cell.

Consider a Fuel Cell where hydrogen fuel (H<sub>2</sub>) is added to the so-called *anode volume*, while oxygen (O<sub>2</sub>) is added to the *cathode volume* (typically via air), where an (acid electrolyte) membrane separates the two volumes and only allows for transport of *proton* (H<sup>+</sup>) from the anode side to the cathode side, Fig. 1.

At the *anode* interface between gas, some catalyst, and the proton conducting membrane (the so-called triple-point junction), hydrogen H<sub>2</sub> reacts in a *half-cell reaction* as follows:




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<sup>1</sup>A H<sub>2</sub> + O<sub>2</sub> mixture is sometimes known as “knallgas” — taken from German and Scandinavian languages = bang-gas

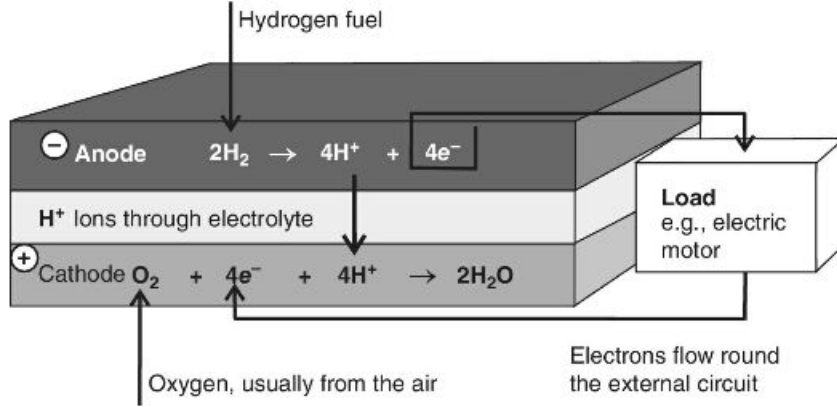


Figure 1: Fuel Cell with acid electrolyte and hydrogen fuel.

Proton  $\text{H}^+$  can move through the membrane towards the *cathode* interface. The *electron*  $\text{e}^-$ , on the other hand, moves through the external electric circuit and eventually ends up at the cathode.

At the *cathode* triple-point junction, oxygen  $\text{O}_2$  reacts with the proton coming through the electrolyte and the electron coming via the external electric circuit in the following *half-cell reaction*:



Combining the two half-cell reactions, the overall reaction is sometimes written as



where the product is *water*, but other possibilities exist.

Let the species be ordered according to the vector  $(\text{H}_2, \text{O}_2, \text{H}_2\text{O})$ . It is of interest to formulate the *stoichiometric matrix*  $\nu$  for the overall reaction in Eq. 3, as well as the *positive stoichiometric parameter* for the *electron*,  $\nu_{\text{e}^-}$  (Note:  $\nu_{\text{e}^-} > 0$ ).

|    | <b>Problem 1:</b> Which of the following pairs of $\nu$ and $\nu_{\text{e}^-}$ is <i>incorrect</i> ? | Choice |
|----|------------------------------------------------------------------------------------------------------|--------|
| A: | $\nu = (1, -\frac{1}{2}, 1)$ and $\nu_{\text{e}^-} = 2$                                              |        |
| B: | $\nu = (-1, -\frac{1}{2}, 1)$ and $\nu_{\text{e}^-} = 2$                                             |        |
| C: | $\nu = (-\frac{1}{2}, -\frac{1}{4}, \frac{1}{2})$ and $\nu_{\text{e}^-} = 1$                         |        |
| D: | $\nu = (-2, -1, 2)$ and $\nu_{\text{e}^-} = 4$                                                       |        |

▲

### Solution 1. Reactions in Fuel Cell.

For Problem 1, combining the half-cell reactions, Eqs. 1–2, to the overall-reaction, Eq. 3, we have

$$\begin{aligned}
 2k (\text{H}_2 - 2\text{H}^+ - 2\text{e}^-) &\rightarrow k (2\text{H}_2\text{O} - \text{O}_2 - 4\text{H}^+ - 4\text{e}^-) \\
 \Downarrow \\
 k \cdot (2\text{H}_2 + \text{O}_2) &\rightarrow k \cdot 2\text{H}_2\text{O} + k \cdot 4 (\text{e}^- - \text{e}^-) + k \cdot 4 (\text{H}^+ - \text{H}^+).
 \end{aligned}$$

From this overall reaction, we can read off the stoichiometric matrix  $\nu$  as

$$\nu = \begin{pmatrix} -2k, & -k, & 2k \end{pmatrix} \quad (4)$$

and corresponding stoichiometric electron parameter

$$\nu_{e-} = 4k. \quad (5)$$

[A similar proton parameter would be  $\nu_{H+} = 4k$ , but we don't need that parameter.]

Correct stoichiometric matrices  $\nu$  and corresponding electron parameters are then any positive choice of  $k$  and corresponding values  $\nu$  and  $\nu_{e-}$  according to Eqs. 4 and 5.

|    | <b>Solution 1:</b> Which of the following pairs of $\nu$ and $\nu_{e-}$ is <i>incorrect</i> ? | Choice |
|----|-----------------------------------------------------------------------------------------------|--------|
| A: | $\nu = (1, -\frac{1}{2}, 1)$ and $\nu_{e-} = 2$                                               | ×      |
| B: | $\nu = (-1, -\frac{1}{2}, 1)$ and $\nu_{e-} = 2$                                              |        |
| C: | $\nu = (-\frac{1}{2}, -\frac{1}{4}, \frac{1}{2})$ and $\nu_{e-} = 1$                          |        |
| D: | $\nu = (-2, -1, 2)$ and $\nu_{e-} = 4$                                                        |        |

▲

### Problem 2. Equilibrium voltage in Fuel Cell

The so-called *chemical potential*  $\mu_j$  of species  $j$  in a mixture is given as molar Gibbs' energy  $\tilde{G}_j^\circ$  of species  $j$  at standard state plus a term  $RT \cdot \ln a_j$  where  $R$  is the gas constant,  $T$  is absolute temperature (i.e., in K), and  $a_j$  is the *activity*. With  $\mu_j \equiv \tilde{G}_j$ , we have

$$\tilde{G}_j = \tilde{G}_j^\circ + RT \cdot \ln a_j. \quad (6)$$

For *ideal gas*,

$$a_j = \frac{p_j}{p^\circ} \quad (7)$$

where  $p_j$  is the partial pressure of species  $j$ , while  $p^\circ$  is a standard state *scaling* pressure (normally either 1 atm or 1 bar). In the liquid case, for an *ideal solution*

$$a_j = x_j \quad (8)$$

where  $x_j$  is the mole fraction  $x_j = n_j/n$  of the liquid.

Next, the molar Gibbs energy of *reaction*  $\Delta_r \tilde{G}$  (at arbitrary  $T$ ,  $p_j$ ) is given as

$$\Delta_r \tilde{G} = \sum_j \nu_j \tilde{G}_j, \quad (9)$$

while the *standard* molar Gibbs energy of reaction (i.e., at arbitrary  $T$  and standard pressure  $p^\circ$ ) is denoted  $\Delta_r \tilde{G}^\circ$ . The equilibrium voltage from the cathode side (positive) to the anode side (negative) is then

$$u_n = -\frac{\Delta_r \tilde{G}}{F \nu_{e-}} \quad (10)$$

where  $F$  is *Faraday's number*.

|    | <b>Problem 2:</b> Equilibrium voltage — which expression is <i>incorrect</i> ?                                         | Choice |
|----|------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $u_n = \left( -\Delta_r \tilde{G}^\circ - RT \sum_j \ln a_j^{\nu_j} \right) / (F \nu_{e-})$                            |        |
| B: | $u_n = \left( -\Delta_r \tilde{G}^\circ - RT \ln \left( \prod_j a_j^{\nu_j} \right) \right) / (F \nu_{e-})$            |        |
| C: | $u_n = \left( -\Delta_r \tilde{G}^\circ - RT \prod_j \ln a_j^{\nu_j} \right) / (F \nu_{e-})$                           |        |
| D: | $u_n = \left( -\Delta_r \tilde{G}^\circ / \nu_{e-} - RT \ln \left( \prod_j a_j^{\nu_j / \nu_{e-}} \right) \right) / F$ |        |

▲

### Solution 2. Equilibrium voltage in Fuel Cell

For Problem 2: by combining Eqs. 10 and 9, we have

$$u_n = -\frac{\sum_j \nu_j \tilde{G}_j}{F\nu_{e^-}} = -\frac{\sum_j \nu_j \left( \tilde{G}_j^\circ + RT \cdot \ln a_j \right)}{F\nu_{e^-}}$$

$$\Downarrow$$

$$u_n = -\frac{\sum_j \nu_j \tilde{G}_j^\circ + \sum_j RT \cdot \nu_j \ln a_j}{F\nu_{e^-}}.$$

Here, we can introduce the standard state voltage  $u_o$  as

$$u_o = -\frac{\sum_j \nu_j \tilde{G}_j^\circ}{F\nu_{e^-}} = -\frac{\Delta_r \tilde{G}^\circ}{F\nu_{e^-}}.$$

Observe also that

$$u_o = -\frac{\sum_j \frac{\nu_j}{\nu_{e^-}} \tilde{G}_j}{F}$$

so that  $u_o$  is independent of the actual choice of  $\nu$  as long as the fraction  $\nu/\nu_{e^-}$  is correct. Here,

$$\frac{\nu}{\nu_{e^-}} = \frac{(-2k, -k, 2k)}{4k} = \left( -\frac{1}{2}, -\frac{1}{4}, \frac{1}{2} \right)$$

is fixed, as it should be.

Next, consider the pressure dependent term  $u_p$ :

$$u_p = -\frac{\sum_j RT \cdot \nu_j \ln a_j}{F\nu_{e^-}} = -\frac{RT}{F\nu_{e^-}} \sum_j \nu_j \ln a_j = -\frac{RT}{F} \sum_j \frac{\nu_j}{\nu_{e^-}} \ln a_j.$$

Here we can use that  $\ln x + \ln y = \ln(x \cdot y)$ , and that  $a \ln x = \ln x^a$ . We then see that we can rephrase  $u_p$  to any of the following

$$u_p = -\frac{RT}{F\nu_{e^-}} \sum_j \nu_j \ln a_j = -\frac{RT}{F\nu_{e^-}} \sum_j \ln a_j^{\nu_j} = -\frac{RT}{F\nu_{e^-}} \ln \left( \prod_j a_j^{\nu_j} \right)$$

or

$$u_p = -\frac{RT}{F} \sum_j \frac{\nu_j}{\nu_{e^-}} \ln a_j = -\frac{RT}{F} \sum_j \ln a_j^{\nu_j/\nu_{e^-}} = -\frac{RT}{F} \ln \left( \prod_j a_j^{\nu_j/\nu_{e^-}} \right).$$

|    | <b>Solution 2:</b> Equilibrium voltage — which expression is <i>incorrect</i> ?                                        | Choice |
|----|------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $u_n = \left( -\Delta_r \tilde{G}^\circ - RT \sum_j \ln a_j^{\nu_j} \right) / (F\nu_{e^-})$                            |        |
| B: | $u_n = \left( -\Delta_r \tilde{G}^\circ - RT \ln \left( \prod_j a_j^{\nu_j} \right) \right) / (F\nu_{e^-})$            |        |
| C: | $u_n = \left( -\Delta_r \tilde{G}^\circ - RT \prod_j \ln a_j^{\nu_j} \right) / (F\nu_{e^-})$                           | ×      |
| D: | $u_n = \left( -\Delta_r \tilde{G}^\circ / \nu_{e^-} - RT \ln \left( \prod_j a_j^{\nu_j/\nu_{e^-}} \right) \right) / F$ |        |

▲

### Problem 3. Gas vs. liquid form of water product.

In the group project, it was assumed that water was present in the cathode volume at *saturation pressure*, which is a pure function of temperature:

$$p_{\text{H}_2\text{O}}^c = p_{\text{H}_2\text{O}}^{\text{sat}}(T).$$

Saturation pressure for water may be given by the Antoine equation,

$$\log_{10} \left( \frac{p_{\text{H}_2\text{O}}^{\text{sat}}}{p^\circ} \right) = A_{\text{H}_2\text{O}} - \frac{B_{\text{H}_2\text{O}}}{T + C_{\text{H}_2\text{O}} - 273.15 \text{ K}} \quad (11)$$

where  $T$  is absolute temperature (K) and

$$\begin{aligned} A_{\text{H}_2\text{O}} &= 5.11564 \\ B_{\text{H}_2\text{O}} &= 1687.537 \text{ K} \\ C_{\text{H}_2\text{O}} &= 230.17 \text{ K}. \end{aligned}$$

This implies that in the expression for  $\tilde{G}_{\text{H}_2\text{O}}$  in Eq. 6, the water activity was set to

$$a_{\text{H}_2\text{O}}^c = \frac{p_{\text{H}_2\text{O}}^{\text{sat}}}{p^\circ}.$$

In much of the relevant published literature, *instead* it is assumed that water is present in the cathode volume as *condensed water*, i.e., water in *liquid form*. Because the other involved species (oxygen, nitrogen, etc.) are (almost) completely in gas form, this implies that the liquid phase is pure water, i.e.,  $x_{\text{H}_2\text{O}}^c \approx 1$ , and the activity of water is unity:

$$a_{\text{H}_2\text{O}}^c = x_{\text{H}_2\text{O}}^c = 1.$$

An interesting question is: *can both assumptions on cathode water state* lead to the same equilibrium voltage, or is the assumption of liquid water leading to  $a_{\text{H}_2\text{O}}^c = 1$  just introduced for convenience?

To help us assess this, under the assumption of *constant heat capacity*  $\tilde{c}_{\text{p},j}$ , we have:

$$\tilde{G}_j^\circ = \Delta_{\text{f}}\tilde{G}_j^\circ - \Delta_{\text{f}}\tilde{S}_j^\circ (T - T^\circ) + \tilde{c}_{\text{p},j} \left( T - T^\circ - T \cdot \ln \frac{T}{T^\circ} \right). \quad (12)$$

For water in *gas phase*:

$$\begin{aligned} \Delta_{\text{f}}\tilde{G}_{\text{H}_2\text{O},\text{g}} &= -228.572 \text{ kJ/mol} \\ \Delta_{\text{f}}\tilde{S}_{\text{H}_2\text{O},\text{g}} &= 188.7 \text{ J/mol K} \\ \tilde{c}_{\text{p},\text{H}_2\text{O},\text{g}} &= 33.58 \text{ J/mol K} \end{aligned}$$

while for water in *liquid phase*:

$$\begin{aligned} \Delta_{\text{f}}\tilde{G}_{\text{H}_2\text{O},\ell} &= -237.129 \text{ kJ/mol} \\ \Delta_{\text{f}}\tilde{S}_{\text{H}_2\text{O},\ell} &= 69.91 \text{ J/mol K} \\ \tilde{c}_{\text{p},\text{H}_2\text{O},\ell} &= 75.29 \text{ J/mol K}. \end{aligned}$$

|    |                                                                                                                                                                                                                                                                                                                            |        |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|    | <b>Problem 3:</b> Assuming H <sub>2</sub> O product in gas or liquid form — which of the following statements is <i>correct</i> ?                                                                                                                                                                                          | Choice |
| A: | The assumption of H <sub>2</sub> O in liquid form is only introduced for <i>convenience</i> , and leads to a slightly different equilibrium voltage $u_n$ compared to assuming gas.                                                                                                                                        |        |
| B: | Whether cathode H <sub>2</sub> O is in liquid or gas form, in both cases we get identical equilibrium voltage $u_n$ .                                                                                                                                                                                                      |        |
| C: | The assumption of H <sub>2</sub> O in gas or liquid form leads to different equilibrium voltages $u_n$ . However, if we take into account the energy balance and <i>condensation</i> of the H <sub>2</sub> O in gas form, this will modify the temperature and will gradually lead to the same equilibrium voltage $u_n$ . |        |
| D: | The H <sub>2</sub> O content on the <i>anode</i> side will also affect the equilibrium voltage $u_n$ , and needs to be taken into account to get a correct equilibrium voltage $u_n$ .                                                                                                                                     |        |

▲

**Solution 3.** Gas vs. liquid form of water product.

For Problem 3,  $u_n$  is given as

$$u_n = -\frac{\Delta_r \tilde{G}}{F\nu_{e^-}} = -\frac{\sum_j \nu_j \tilde{G}_j}{F\nu_{e^-}}.$$

From this expression, we see that the question is only related to the phase of H<sub>2</sub>O, and that it is sufficient to consider  $\tilde{G}_{\text{H}_2\text{O}}$ ,

$$\tilde{G}_{\text{H}_2\text{O}} = \tilde{G}_{\text{H}_2\text{O}}^\circ + RT \ln a_{\text{H}_2\text{O}}.$$

So the question is to compare  $\tilde{G}_{\text{H}_2\text{O}}$  where  $\tilde{G}_{\text{H}_2\text{O},\ell}^\circ$  is computed from Eq. 12 with  $\Delta_f \tilde{G}_{\text{H}_2\text{O},\ell}^\circ$ ,  $\Delta_f \tilde{S}_{\text{H}_2\text{O},\ell}$ , and  $\tilde{c}_{\text{p,H}_2\text{O},\ell}$ , and  $a_{\text{H}_2\text{O},\ell} = x_{\text{H}_2\text{O},\ell} = 1$ , vs.  $\tilde{G}_{\text{H}_2\text{O},\text{g}}^\circ$  computed from Eq. 12 with  $\Delta_f \tilde{G}_{\text{H}_2\text{O},\text{g}}^\circ$ ,  $\Delta_f \tilde{S}_{\text{H}_2\text{O},\text{g}}^\circ$ , and  $\tilde{c}_{\text{p,H}_2\text{O},\text{g}}^\circ$ , and  $a_{\text{H}_2\text{O},\text{g}} = \frac{p_{\text{H}_2\text{O}}}{p^\circ}$  with  $p_{\text{H}_2\text{O}} = p_{\text{H}_2\text{O}}^{\text{sat}}(T)$  with  $p_{\text{H}_2\text{O}}^{\text{sat}}$  computed from Eq. 11. The result is as shown in Fig. 2.

Observe that although  $\tilde{G}_{\text{H}_2\text{O},\text{g}}^\circ \neq \tilde{G}_{\text{H}_2\text{O},\ell}^\circ$ , when we add the activity correction which is  $RT \ln \frac{p_{\text{H}_2\text{O}}^{\text{sat}}}{p^\circ}$  for gas and  $RT \ln 1 = 0$  for liquid, the two molar Gibbs energies of water are identical. Because of this, we will then also get identical equilibrium voltages  $u_n$ .

|    |                                                                                                                                                                                                                                                                                                                            |        |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|    | <b>Solution 3:</b> Assuming H <sub>2</sub> O product in gas or liquid form — which of the following statements is <i>correct</i> ?                                                                                                                                                                                         | Choice |
| A: | The assumption of H <sub>2</sub> O in liquid form is only introduced for <i>convenience</i> , and leads to a slightly different equilibrium voltage $u_n$ compared to assuming gas.                                                                                                                                        |        |
| B: | Whether cathode H <sub>2</sub> O is in liquid or gas form, in both cases we get identical equilibrium voltage $u_n$ .                                                                                                                                                                                                      | ×      |
| C: | The assumption of H <sub>2</sub> O in gas or liquid form leads to different equilibrium voltages $u_n$ . However, if we take into account the energy balance and <i>condensation</i> of the H <sub>2</sub> O in gas form, this will modify the temperature and will gradually lead to the same equilibrium voltage $u_n$ . |        |
| D: | The H <sub>2</sub> O content on the <i>anode</i> side will also affect the equilibrium voltage $u_n$ , and needs to be taken into account to get a correct equilibrium voltage $u_n$ .                                                                                                                                     |        |

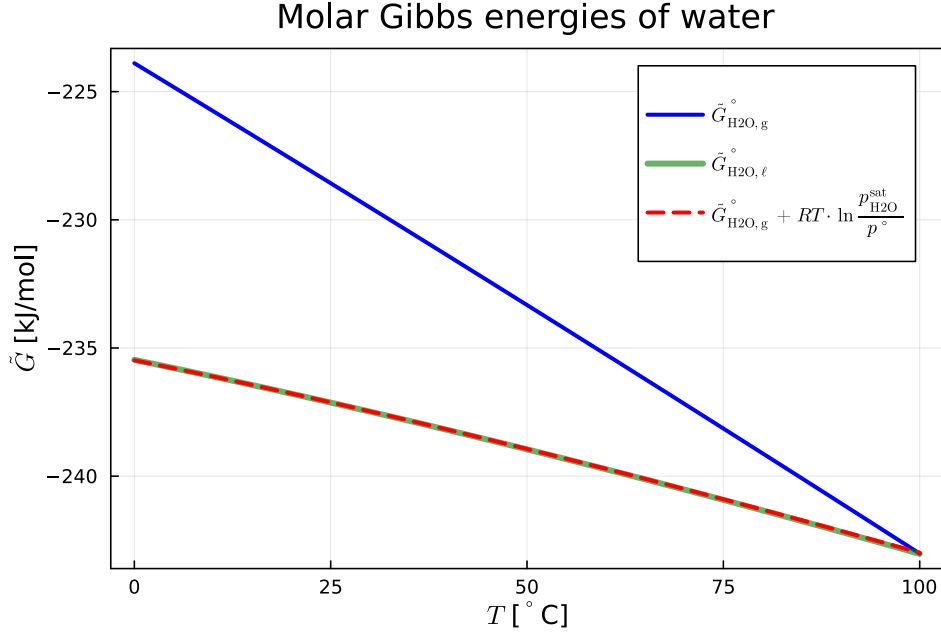


Figure 2: Molar Gibbs energy of water.

▲

**Problem 4.** Pressure effect on equilibrium voltage.

Assuming that water is produced in *gas phase*, the standard state equilibrium voltage  $u_o$  is as in Fig. 3.

If any of the partial pressures  $p_{H_2}$ ,  $p_{O_2}$ , or  $p_{H_2O}$  deviate from the standard state  $p^\circ = 1$  bar, there will be a correction term to the equilibrium voltage as indicated in Problem 2. We will assume that cathode water is at the saturation pressure (water is produced at the cathode), which for  $T = 80^\circ\text{C}$  is  $p_{H_2O}^c \approx p_{H_2O}^{\text{sat}} \approx 0.4$  bar, while the anode water pressure is  $p_{H_2O}^a \approx \frac{1}{2}p_{H_2O}^{\text{sat}} \approx 0.2$  bar (the membrane must contain some water to work, and some water is normally added to the anode volume).

For PEM fuel cells, the total anode pressure  $p_a$  and total cathode pressure  $p_c$  should be equal, and not deviate by more than  $\Delta p_{ac} = |p_a - p_c| \leq 0.2$  bar to avoid mechanical damage to the membrane. Typical electrode pressures  $p_a$ ,  $p_c$  lie in the range of  $[1, 5]$  bar.

Assume that we operate the PEM fuel cell with electrode pressures  $p_a = p_c = 5$  bar, that we feed the anode with pure hydrogen (+ some water), while we feed the cathode with air (+ some water) where the oxygen mole fraction is approximately 0.2. Assume that we feed sufficient air to the cathode side to maintain mole fraction oxygen of 0.2 in the cathode volume.

What is the approximate change  $u_p$  in equilibrium voltage away from the standard state voltage ( $u_o$ ; Fig. 3) due to pressure conditions as indicated above? Use gas constant  $R = 8.31 \text{ J}/(\text{mol} \cdot \text{K})$ , Faraday's constant  $F = 9.648 \cdot 10^4 \text{ C/mol}$ , and assume operation at  $T$  given by  $80^\circ\text{C}$ .

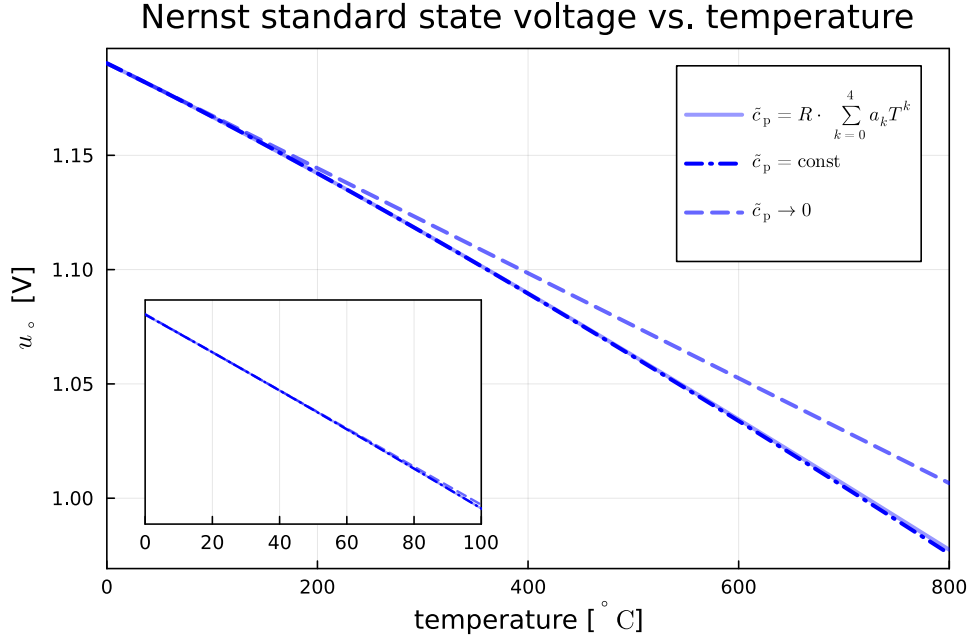


Figure 3: Hydrogen Fuel Cell standard state equilibrium voltage  $u_o$ .

|    |                                                                                                                 |        |
|----|-----------------------------------------------------------------------------------------------------------------|--------|
|    | <b>Problem 4:</b> Approximate change $u_p$ in equilibrium voltage due to pressure conditions specified above is | Choice |
| A: | $u_p \approx 0.12 \text{ V}$                                                                                    |        |
| B: | $u_p \approx -33 \text{ mV}$                                                                                    |        |
| C: | $u_p \approx 0.31 \text{ V}$                                                                                    |        |
| D: | $u_p \approx 37 \text{ mV}$                                                                                     |        |



**Solution 4.** Pressure effect on equilibrium voltage.

For Problem 4: with the specified pressure data,

$$p_{\text{H}_2} = p_a - p_{\text{H}_2\text{O}}^a = 5 - 0.2 = 4.8 \text{ bar}$$

$$p_{\text{O}_2} = p_c - p_{\text{H}_2\text{O}}^c - p_{\text{N}_2}.$$

With “infinite” flow rate of air through the cathode volume,  $p_{\text{N}_2} = 4p_{\text{O}_2}$ ; with finite flow,  $p_{\text{N}_2} > 4p_{\text{O}_2}$  (since oxygen is consumed in the reaction). Assuming infinite flow rate, we have

$$p_{\text{O}_2} = p_c - p_{\text{H}_2\text{O}}^c - 4p_{\text{O}_2}$$

$$\Downarrow$$

$$p_{\text{O}_2} = \frac{1}{5} (p_c - p_{\text{H}_2\text{O}}^c) = \frac{1}{5} (5 - 0.4) = 1 - 0.08 = 0.92 \text{ bar}.$$

For finite flow rate,  $p_{\text{O}_2}$  will be smaller than 0.92 bar.

We find that the correction voltage  $u_p$  is

$$u_p = \frac{RT}{F\nu_{e^-}} \ln \left( \frac{\left( \frac{p_{\text{H}_2}}{p^\circ} \right) \sqrt{\frac{p_{\text{O}_2}}{p^\circ}}}{\left( \frac{p_{\text{H}_2\text{O}}^c}{p^\circ} \right)} \right).$$

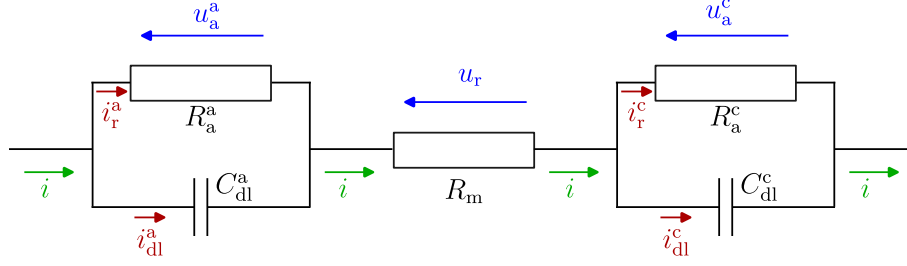


Figure 4: Equivalent linear electric model for kinetics of membrane.

Here,  $\nu_{e-} = 2$  when  $\nu = (-1, -\frac{1}{2}, 1)$  (as used above). We thus find

$$u_p = \frac{8.31 \cdot (80 + 273.15)}{9.648 \cdot 10^4 \cdot 2} \ln \left( \frac{4.8 \cdot \sqrt{0.92}}{0.4} \right) \approx 0.037 \text{ V} = 37 \text{ mV}.$$

|    | <b>Solution 4:</b> Approximate change $u_p$ in equilibrium voltage due to pressure conditions specified above is | Choice |
|----|------------------------------------------------------------------------------------------------------------------|--------|
| A: | $u_p \approx 0.12 \text{ V}$                                                                                     |        |
| B: | $u_p \approx -33 \text{ mV}$                                                                                     |        |
| C: | $u_p \approx 0.31 \text{ V}$                                                                                     |        |
| D: | $u_p \approx 37 \text{ mV}$                                                                                      | ×      |

**Problem 5.** Transfer function for Activation voltage loss.

On the convoluted interface between a charged eletrode and the electrolyte, there will be a concentration of ions in the electrolyte with opposite charge as to that of the electrode. This creates a *capacitance*, and there will be such a capacitance on both electrodes. In general, capacitance is given by

$$C_{dl} = \varepsilon_{dl} \frac{A_{dl}}{\delta_{dl}} \quad (13)$$

where  $\varepsilon_{dl}$  is the *permittivity* of the space between the ions in the double layer,  $A_{dl}$  is the area of the double layer, and  $\delta_{dl}$  is the separation distance between the ions in the double layer. Due to large area  $A_{dl}$  and short distance  $\delta_{dl}$ , the capacitance  $C_{dl}$  can be quite large, and in the order of a few Farads. Theoretical expressions exist for  $C_{dl}$ .

In the group project, activation voltage loss was only considered on *one* electrode. Figure 4 shows a linearized electric circuit equivalent of the membrane kinetics, with external current  $i$ , activation resistances  $R_a^a$ ,  $R_a^c$  at the anode and cathode electrode, respectively (given by linearizing the Butler-Volmer equation, or similar), double layer capacitors  $C_{dl}^a$ ,  $C_{dl}^c$  at the anode and cathode electrodes, respectively, and membrane resistance  $R_m$ . Figure 4 shows the relevant currents  $i$  (green color),  $i_r^a$ ,  $i_{dl}^a$ ,  $i_r^c$ , and  $i_{dl}^c$  (red color), ans well as relevant voltages  $u_a^a$ ,  $u_r$ , and  $u_a^c$  (blue color). Because the depicted circuit in Fig. 4 is linear, the order of the blocks is irrelevant, and we will neglect the membrane resistor  $R_m$  in the sequel; we will set the *total* activation voltage  $u_a$  equal to the *sum* of activation voltage losses on the anode ( $u_a^a$ ) and the cathode ( $u_a^c$ ). Because the course syllabus for course FM1015 Modelling of Dynamic Systems does not include modeling of electric circuits, the model for the activation voltages is given below for

convenience:

$$C_{dl}^a \frac{du_a^a}{dt} = i_{dl}^a \quad (14)$$

$$u_a^a = R_a^a i_r^a \quad (15)$$

$$i = i_r^a + i_{dl}^a \quad (16)$$

$$C_{dl}^c \frac{du_a^c}{dt} = i_{dl}^c \quad (17)$$

$$u_a^c = R_a^c i_r^c \quad (18)$$

$$i = i_r^c + i_{dl}^c \quad (19)$$

$$u_a = u_a^a + u_a^c. \quad (20)$$

It is of interest to find the *transfer function*  $h(s)$  from external current  $i$  to total activation voltage loss  $u_a$ , i.e.,

$$u_a(s) = h(s) \cdot i(s).$$

|    | <b>Problem 5:</b> Which expression for the transfer function from $i(s)$ to $u_a(s)$ , denoted $h(s)$ , is <i>correct</i> ? | Choice |
|----|-----------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $h(s) = \frac{R_a^a}{1+R_a^a C_{dl}^a s} \cdot \frac{R_a^c}{1+R_a^c C_{dl}^c s}$                                            |        |
| B: | $h(s) = (R_a^a + R_a^c) \frac{1+(C_{dl}^a + C_{dl}^c)s}{(1+R_a^a C_{dl}^a s)(1+R_a^c C_{dl}^c s)}$                          |        |
| C: | $h(s) = (C_{dl}^a + C_{dl}^c) \frac{R_a^a R_a^c + (R_a^a + R_a^c)s}{(1+R_a^a C_{dl}^a s)(1+R_a^c C_{dl}^c s)}$              |        |
| D: | $h(s) = \frac{R_a^a}{1+R_a^a C_{dl}^a s} + \frac{R_a^c}{1+R_a^c C_{dl}^c s}$                                                |        |

▲

**Solution 5.** Transfer function for Activation voltage loss.

For Problem 5: the transfer function from  $i$  to  $u_a^a$  is

$$\begin{aligned}
 sC_{dl}^a u_a^a &= i_{dl}^a = i - i_r^a = i - \frac{u_a^a}{R_a^a} \\
 &\Downarrow \\
 (1 + R_a^a C_{dl}^a s) u_a^a &= R_a^a i \\
 &\Downarrow \\
 u_a^a &= \frac{R_a^a}{1 + R_a^a C_{dl}^a s} i.
 \end{aligned}$$

Similarly, we will find

$$u_a^c = \frac{R_a^c}{1 + R_a^c C_{dl}^c s} i.$$

The total activation voltage is

$$\begin{aligned}
 u_a &= u_a^a + u_a^c \\
 &= \Downarrow \\
 u_a &= \underbrace{\left( \frac{R_a^a}{1 + R_a^a C_{dl}^a s} + \frac{R_a^c}{1 + R_a^c C_{dl}^c s} \right)}_{h(s)} i.
 \end{aligned}$$

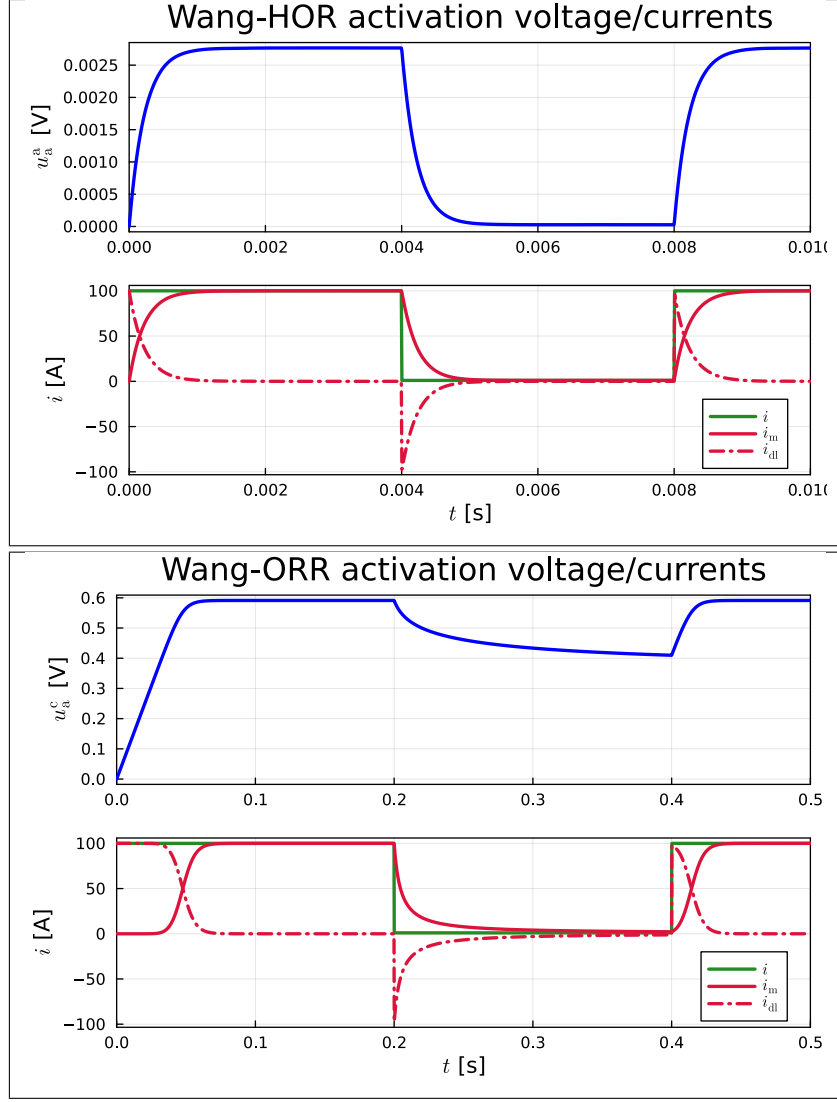


Figure 5: Step responses in activation voltages at the anode ( $u_a^a$ ) and cathode ( $u_a^c$ ), respectively.

|    | <b>Solution 5:</b> Which expression for the transfer function from $i(s)$ to $u_a(s)$ , denoted $h(s)$ , is <b>correct</b> ? | Choice |
|----|------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $h(s) = \frac{R_a^a}{1+R_a^a C_{dl}^a s} \cdot \frac{R_a^c}{1+R_a^c C_{dl}^c s}$                                             |        |
| B: | $h(s) = (R_a^a + R_a^c) \frac{1+(C_{dl}^a + C_{dl}^c)s}{(1+R_a^a C_{dl}^a s)(1+R_a^c C_{dl}^c s)}$                           |        |
| C: | $h(s) = (C_{dl}^a + C_{dl}^c) \frac{R_a^a R_a^c + (R_a^a + R_a^c)s}{(1+R_a^a C_{dl}^a s)(1+R_a^c C_{dl}^c s)}$               |        |
| D: | $h(s) = \frac{R_a^a}{1+R_a^a C_{dl}^a s} + \frac{R_a^c}{1+R_a^c C_{dl}^c s}$                                                 | ×      |

**Problem 6.** Double layer capacitance.

Using a more accurate but more complex model than the Butler-Volmer equation, step changes in external current  $i$  leads to the responses in activation voltage losses  $u_a^a$  (Wang-HOR) and  $u_a^c$  (Wang-ORR) as depicted in Fig. 5. In particular, observe (i) the differences in activation voltage losses at the electrodes, and (ii) the differences in time constants.

The transfer function in Problem 5 has *two* time constants,  $T_a = R_a^a C_{dl}^a$  and  $T_c = R_a^c C_{dl}^c$  at the anode and cathode, respectively. In the sequel, assume for simplicity that  $C_{dl}^a = C_{dl}^c = C_{dl}$ .

Given the large difference in gain and time constants for the transfer functions, propose a *simplified* transfer function  $h_{\text{simp}}(s)$  for the activation voltage loss, and suggest the corresponding simplified capacitor  $C_{\text{simp}}$  to use.

|    | <b>Problem 6:</b> Simplified transfer function $h_{\text{simp}}(s)$ with overall capacitor $C_{\text{simp}}$ — which of the following pairs is <i>correct</i> ? | Choice |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $h_{\text{simp}}(s) = R_a^c / (1 + R_a^c C_{\text{simp}} s)$ , $C_{\text{simp}} = C_{dl}/2$                                                                     |        |
| B: | $h_{\text{simp}}(s) = R_a^a / (1 + R_a^a C_{\text{simp}} s)$ , $C_{\text{simp}} = C_{dl}/2$                                                                     |        |
| C: | $h_{\text{simp}}(s) = R_a^c / (1 + R_a^c C_{\text{simp}} s)$ , $C_{\text{simp}} = C_{dl}$                                                                       |        |
| D: | $h_{\text{simp}}(s) = R_a^a / (1 + R_a^a C_{\text{simp}} s)$ , $C_{\text{simp}} = C_{dl}$                                                                       |        |

▲

**Solution 6.** Double layer capacitance.

For Problem 6, we observe from Fig. 5 that  $T_a \ll T_c$ . By assuming that  $C_{dl}^a \approx C_{dl}^c = C_{dl}$ , this means that  $R_a^a \ll R_a^c$ . From the correct transfer function from  $i$  to  $u_a$ ,

$$h(s) = \frac{R_a^a}{1 + R_a^a C_{dl}^a s} + \frac{R_a^c}{1 + R_a^c C_{dl}^c s}$$

and we let  $R_a^a \rightarrow 0$  leading to

$$h_{\text{simp}}(s) = \frac{R_a^c}{1 + R_a^c C_{dl}^c s} = \frac{R_a^c}{1 + R_a^c C_{dl} s}.$$

We see that  $C_{\text{simp}} = C_{dl}$ .

This result is interesting, in that some of the fuel cell literature claims that the double layer capacitors can be considered two equal capacitors in series, leading to a resulting capacitance equal to half of the double-layer capacitance, i.e., some literature claims that

$$C_{\text{simp}} = \frac{C_{dl}}{2}$$

— which we have shown is false. The reason for this misunderstanding is that the capacitors are *not* in true series with each others — instead, they are two RC circuits in series.

|    | <b>Solution 6:</b> Simplified transfer function $h_{\text{simp}}(s)$ with overall capacitor $C_{\text{simp}}$ — which of the following pairs is <i>correct</i> ? | Choice |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $h_{\text{simp}}(s) = R_a^c / (1 + R_a^c C_{\text{simp}} s)$ , $C_{\text{simp}} = C_{dl}/2$                                                                      |        |
| B: | $h_{\text{simp}}(s) = R_a^a / (1 + R_a^a C_{\text{simp}} s)$ , $C_{\text{simp}} = C_{dl}/2$                                                                      |        |
| C: | $h_{\text{simp}}(s) = R_a^c / (1 + R_a^c C_{\text{simp}} s)$ , $C_{\text{simp}} = C_{dl}$                                                                        | ×      |
| D: | $h_{\text{simp}}(s) = R_a^a / (1 + R_a^a C_{\text{simp}} s)$ , $C_{\text{simp}} = C_{dl}$                                                                        |        |

▲

**Problem 7.** Electric time constants.

Based on the two plots in Fig. 5, what are approximate values for time constants in the two plots (see  $u_a$  in the Wang-HOR plot and  $u_a$  in the Wang-ORR plot)?

|    | <b>Problem 7:</b> Approximate electric time constants for the anode and the cathode, which pair is <i>reasonable</i> ? | Choice |
|----|------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $T_a \approx 0.5 \text{ ms}$ , $T_c \approx 0.1 \text{ s}$                                                             |        |
| B: | $T_a \approx 2 \text{ ms}$ , $T_c \approx 10 \text{ s}$                                                                |        |
| C: | $T_a \approx 0.01 \text{ s}$ , $T_c \approx 0.5 \text{ s}$                                                             |        |
| D: | $T_a \approx 100 \text{ ms}$ , $T_c \approx 2 \text{ s}$                                                               |        |



**Solution 7.** Electric time constants.

For Problem 7: Fig. 5 shows that the dynamics for  $u_a^a$  operates in a relatively *linear* regime, and with some degree of certainty we can identify a time constant  $T_a \approx 0.0005 \text{ s} = 0.5 \text{ ms}$ . At the same time, the dynamics for  $u_a^c$  is extremely non-linear, where the idea of time constant is questionable. However, pretending the the dynamics in the step at  $t = 0.2 \text{ s}$  is “quasi-linear”, using the 63% rule we can identify a time constant of (very) approximately  $T_c \approx 0.5 \text{ s}$ .

|    | <b>Solution 7:</b> Approximate electric time constants for the anode and the cathode, which pair is <i>reasonable</i> ? | Choice |
|----|-------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $T_a \approx 0.5 \text{ ms}, T_c \approx 0.1 \text{ s}$                                                                 | ×      |
| B: | $T_a \approx 2 \text{ ms}, T_c \approx 10 \text{ s}$                                                                    |        |
| C: | $T_a \approx 0.01 \text{ s}, T_c \approx 0.5 \text{ s}$                                                                 |        |
| D: | $T_a \approx 100 \text{ ms}, T_c \approx 2 \text{ s}$                                                                   |        |



**Problem 8.** Anode species balance.

Figure 6 illustrates the most common gas subsystem for a fuel cell: hydrogen is supplied from a high pressure storage system via an inlet anode pressure controller, alternatively a flow controller. Effluent hydrogen from the anode volume is returned to the inlet via a fan (low anode pressure) or compressor (high anode pressure). The purge valve is normally closed, but can be opened for a short time to purge out accumulated impurities in the hydrogen gas.

Oxygen is normally supplied via atmospheric air (approximately 20% oxygen, 80% nitrogen), either by fan (low cathode pressure) or compressor (high cathode pressure). The fan/compressor flow rate can be controlled (speed control), and the effluent volume is controlled to pressure  $p_c^e$  (so-called back-pressure controller). If pure oxygen were supplied from a high pressure storage tank (conceptually possible), the oxygen feed system would have to be similar to the hydrogen supply system in the figure.

Assume that the well mixed anode volume  $V_a$  contains only  $\text{H}_2\text{O} + \text{H}_2$ , and that the water content is fixed to be  $p_{\text{H}_2\text{O}}^a = \frac{1}{2}p_{\text{H}_2\text{O}}^{\text{sat}}(T)$  where the saturation pressure is given by Antoine’s model as in Problem. 3.

Assuming ideal gas and constant temperature, and known influent  $\dot{n}_{\text{H}_2}^i$ , the hydrogen species balance leads to the following model for hydrogen pressure:

|    | <b>Problem 8:</b> Dynamic model for hydrogen pressure under given conditions — which model is <i>correct</i> ?                                        | Choice |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\frac{dp_{\text{H}_2}}{dt} = \frac{p_{\text{H}_2}}{V_a} \dot{V}_a^e - \frac{RT}{V_a} \dot{n}_{\text{H}_2}^i - \frac{RT}{V_a} \dot{n}_{\text{H}_2}^g$ |        |
| B: | $\frac{dp_{\text{H}_2}}{dt} = \frac{RT}{V_a} \dot{n}_{\text{H}_2}^i - \frac{p_{\text{H}_2}}{V_a} \dot{V}_a^e + \frac{RT}{V_a} \dot{n}_{\text{H}_2}^g$ |        |
| C: | $\frac{dp_{\text{H}_2}}{dt} = \frac{RT}{V_a} \dot{n}_{\text{H}_2}^i - \frac{p_{\text{H}_2}}{V_a} \dot{V}_a^e$                                         |        |
| D: | $\frac{dp_{\text{H}_2}}{dt} = \frac{p_{\text{H}_2}}{V_a} \dot{V}_a^i + \frac{V_a}{RT} \dot{n}_{\text{H}_2}^i - \frac{RT}{V_a} \dot{n}_{\text{H}_2}^g$ |        |



**Solution 8.** Anode species balance.

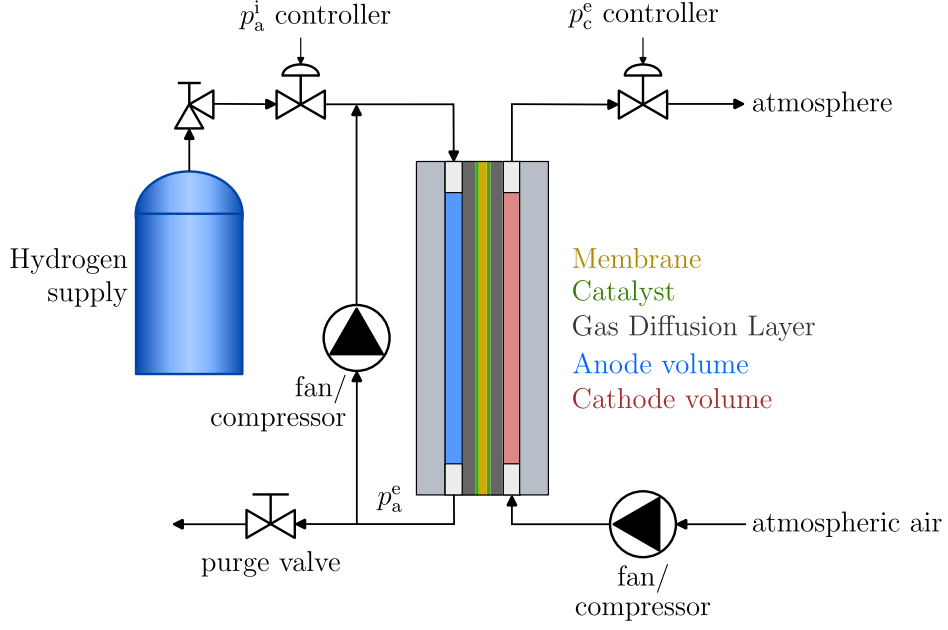


Figure 6: Gas subsystem for membrane.

For Problem 8: The hydrogen species balance is

$$\frac{dn_{H_2}}{dt} = \dot{n}_{H_2}^i - \dot{n}_{H_2}^e + \dot{n}_{H_2}^g.$$

With

$$p_{H_2} V_a = n_{H_2} RT$$

$$p_{H_2} \dot{V}_a^e = \dot{n}_{H_2}^e RT$$

we find

$$\frac{dp_{H_2}}{dt} = \frac{RT}{V_a} \left( \dot{n}_{H_2}^i - \frac{p_{H_2} \dot{V}_a^e}{RT} + \dot{n}_{H_2}^g \right)$$

$$\Downarrow$$

$$\frac{dp_{H_2}}{dt} = \frac{RT}{V_a} \dot{n}_{H_2}^i - \frac{p_{H_2}}{V_a} \dot{V}_a^e + \frac{RT}{V_a} \dot{n}_{H_2}^g.$$

|    | <b>Solution 8:</b> Dynamic model for hydrogen pressure under given conditions — which model is <i>correct</i> ?           | Choice |
|----|---------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\frac{dp_{H_2}}{dt} = \frac{p_{H_2}}{V_a} \dot{V}_a^e - \frac{RT}{V_a} \dot{n}_{H_2}^i - \frac{RT}{V_a} \dot{n}_{H_2}^g$ |        |
| B: | $\frac{dp_{H_2}}{dt} = \frac{RT}{V_a} \dot{n}_{H_2}^i - \frac{p_{H_2}}{V_a} \dot{V}_a^e + \frac{RT}{V_a} \dot{n}_{H_2}^g$ | ×      |
| C: | $\frac{dp_{H_2}}{dt} = \frac{RT}{V_a} \dot{n}_{H_2}^i - \frac{p_{H_2}}{V_a} \dot{V}_a^e$                                  |        |
| D: | $\frac{dp_{H_2}}{dt} = \frac{p_{H_2}}{V_a} \dot{V}_a^i + \frac{V_a}{RT} \dot{n}_{H_2}^i - \frac{RT}{V_a} \dot{n}_{H_2}^g$ |        |

▲

**Problem 9.** Anode effluent flow rate.

Hydrogen is a precious fuel, and should not be wasted. For the gas subsystem in Fig. 6, assume for simplicity that  $\dot{n}_{\text{H}_2}^i$  is a known/controlled function of time. Assume furthermore that the inlet pressure  $p_a^i$  to the anode volume is known and constant, and that a fan or compressor manages to keep a constant pressure increase in the *recycle* flow so that  $\Delta p_a^{\text{rc}} = p_a^i - p_a^e$  is constant. From Bernoulli's steady state energy balance, a typical valve model for the effluent flow rate would be

$$\dot{V}_a^e = K_a \cdot \sqrt{\frac{\Delta p_a^e}{p^o}}. \quad (21)$$

Consider the term  $\dot{V}_a^e$  in Problem 8. Modify the model in Eq. 21 so that (i) the flow unidirectional (i.e., only can flow out), and (ii) give expressions for  $\Delta p_a^e$ .

|    | <b>Problem 9:</b> Modified anode effluent flow rate $\dot{V}_a^e$ — which set of expressions is <i>correct</i> ?                                                                           | Choice |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\dot{V}_a^e = K_a \cdot \sqrt{\max(\Delta p_a^e, 0) / p^o}$ ; $\Delta p_a^e = p_a - p_a^i$ ; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$                                            |        |
| B: | $\dot{V}_a^e = K_a \cdot \sqrt{\min(\Delta p_a^e, 0) / p^o}$ ; $\Delta p_a^e = p_a - p_a^i$ ; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$                                            |        |
| C: | $\dot{V}_a^e = K_a \cdot \sqrt{\max(\Delta p_a^e, 0) / p^o}$ ; $\Delta p_a^e = p_a - p_a^e$ ; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$ ; $p_a^e = p_a^i - \Delta p_a^{\text{rc}}$ |        |
| D: | $\dot{V}_a^e = K_a \cdot \sqrt{\min(\Delta p_a^e, 0) / p^o}$ ; $\Delta p_a^e = p_a - p_a^e$ ; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$ ; $p_a^e = p_a^i - \Delta p_a^{\text{rc}}$ |        |

▲

**Solution 9.** Anode effluent flow rate.

For Problem 9: this is relatively straightforward...

|    | <b>Solution 9:</b> Modified anode effluent flow rate $\dot{V}_a^e$ — which set of expressions is <i>correct</i> ?                                                                          | Choice |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\dot{V}_a^e = K_a \cdot \sqrt{\max(\Delta p_a^e, 0) / p^o}$ ; $\Delta p_a^e = p_a - p_a^i$ ; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$                                            |        |
| B: | $\dot{V}_a^e = K_a \cdot \sqrt{\min(\Delta p_a^e, 0) / p^o}$ ; $\Delta p_a^e = p_a - p_a^i$ ; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$                                            |        |
| C: | $\dot{V}_a^e = K_a \cdot \sqrt{\max(\Delta p_a^e, 0) / p^o}$ ; $\Delta p_a^e = p_a - p_a^e$ ; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$ ; $p_a^e = p_a^i - \Delta p_a^{\text{rc}}$ | ×      |
| D: | $\dot{V}_a^e = K_a \cdot \sqrt{\min(\Delta p_a^e, 0) / p^o}$ ; $\Delta p_a^e = p_a - p_a^e$ ; $p_a = p_{\text{H}_2} + p_{\text{H}_2\text{O}}^a$ ; $p_a^e = p_a^i - \Delta p_a^{\text{rc}}$ |        |

▲

**Problem 10.** Anode hydrogen generation.

Consider the term  $\dot{n}_{\text{H}_2}^g$  in Problem 8. With  $\nu = (\nu_{\text{H}_2}, \nu_{\text{O}_2}, \nu_{\text{H}_2\text{O}})$ , what is the correct expression for  $\dot{n}_{\text{H}_2}^g$  for a single fuel cell?

|    | <b>Problem 10:</b> Which of the following expressions for hydrogen generation $\dot{n}_{\text{H}_2}^g$ is <i>correct</i> ? | Choice |
|----|----------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\dot{n}_{\text{H}_2}^g = \nu_{\text{H}_2} \cdot i / F$                                                                    |        |
| B: | $\dot{n}_{\text{H}_2}^g = -\nu_{\text{H}_2} \cdot i / F$                                                                   |        |
| C: | $\dot{n}_{\text{H}_2}^g = \nu_{\text{H}_2} \cdot i / (F \nu_{e-})$                                                         |        |
| D: | $\dot{n}_{\text{H}_2}^g = -\nu_{\text{H}_2} \cdot i / (F \nu_{e-})$                                                        |        |

▲

**Solution 10.** Anode hydrogen generation.

For Problem 10: current  $i$  divided by Faraday's constant  $F$  gives the flow rate of electrons  $\dot{n}_{e^-}$ ,

$$\dot{n}_{e^-} = \frac{i}{F}.$$

To convert from electrons to hydrogen, we need to include the number of hydrogen molecules consumed per number of electrons involved, i.e.,  $\nu_{H_2}/\nu_{e^-}$ . In total:

$$\dot{n}_{H_2}^g = \frac{\nu_{H_2}}{\nu_{e^-}} \dot{n}_{e^-} = \frac{\nu_{H_2}}{F \nu_{e^-}} i.$$

Because we have made sure that  $\nu_{H_2}$  is negative while  $\nu_{e^-}$  is positive, this ensures that the “generation” term is negative.

|    | <b>Solution 10:</b> Which of the following expressions for hydrogen generation $\dot{n}_{H_2}^g$ is <i>correct</i> ? | Choice |
|----|----------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\dot{n}_{H_2}^g = \nu_{H_2} \cdot i / F$                                                                            |        |
| B: | $\dot{n}_{H_2}^g = -\nu_{H_2} \cdot i / F$                                                                           |        |
| C: | $\dot{n}_{H_2}^g = \nu_{H_2} \cdot i / (F \nu_{e^-})$                                                                | ×      |
| D: | $\dot{n}_{H_2}^g = \nu_{H_2} \cdot i / F$                                                                            |        |

▲

### Problem 11. Anode diffusion.

There is a possible gas pressure drop from the anode volume to the surface of the catalyst/membrane due to diffusion resistance through the Gas Diffusion Layer (Fig. 6), etc. With only water vapor and hydrogen in the anode volume, a possible expression for the resulting pressure drop in hydrogen is

$$\frac{p_{H_2}^{as}}{p_{H_2}^a} = \frac{p_a - p_{H_2O}^a \cdot \exp(f_a(T) \cdot i'')}{p_a - p_{H_2O}^a} \quad (22)$$

where superscripts a and as refer to *bulk* anode volume and *anode surface* (i.e., at the catalyst/membrane surface),  $p_{H_2O}^a = \frac{1}{2} p_{H_2O}^{sat}(T)$ ,  $f_a(T) > 0$  is a known function of temperature (involving diffusion coefficients, etc.), while  $i''$  is the current density in the membrane.

What modifications to the model is required if this diffusion term in Eq. 22 is important?

|    | <b>Problem 11:</b> Which of the following statements about the effect of anode diffusion is <i>correct</i> ?                                                           | Choice |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | Diffusion drop in pressure has no effect on the model: a reduction in hydrogen pressure results in an increase in the water pressure, and these changes cancel out.    |        |
| B: | The anode volume model is as found above in Problems 8–10. However, the pressure $p_{H_2}^{as}$ should be used in the electric model to compute the fuel cell voltage. |        |
| C: | The partial pressure of hydrogen in Problem 8 must be changed to $p_{H_2}^{as}$ .                                                                                      |        |
| D: | It is not necessary to modify the model because the controller for hydrogen feed will cancel out the effect.                                                           |        |

▲

### Solution 11. Anode diffusion.

For Problem 11: the electric part of the model depends on the concentrations at the membrane/catalyst surface. We must therefore use the model for hydrogen pressure in the anode volume as in Problems 8–10; the resulting partial pressure for hydrogen is  $p_{\text{H}_2}^a$ . Then we must compute the hydrogen partial pressure  $p_{\text{H}_2}^{\text{as}}$  at the *anode* catalyst/membrane *surface* using Eq. 22, where  $p_{\text{H}_2\text{O}}^a = \frac{1}{2}p_{\text{H}_2\text{O}}^{\text{sat}}(T)$ , and then use  $p_{\text{H}_2}^{\text{as}}$  in the expression for the activity  $a_{\text{H}_2} = p_{\text{H}_2}^{\text{as}}/p^\circ$  in the pressure dependent voltage  $u_p$ .

|    | <b>Solution 11:</b> Which of the following statements about the effect of anode diffusion is <i>correct</i> ?                                                                        | Choice |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | Diffusion drop in pressure has no effect on the model: a reduction in hydrogen pressure results in an increase in the water pressure, and these changes cancel out.                  |        |
| B: | The anode volume model is as found above in Problems 8–10. However, the pressure $p_{\text{H}_2}^{\text{as}}$ should be used in the electric model to compute the fuel cell voltage. | ×      |
| C: | The partial pressure of hydrogen in Problem 8 must be changed to $p_{\text{H}_2}^{\text{as}}$ .                                                                                      |        |
| D: | It is not necessary to modify the model because the controller for hydrogen feed will cancel out the effect.                                                                         |        |

▲

### Problem 12. Cathode oxygen balance.

Assume that the well mixed cathode volume  $V_c$  contains  $\text{H}_2\text{O} + \text{O}_2 + \text{N}_2$ , and that the water content is fixed to be  $p_{\text{H}_2\text{O}}^c = p_{\text{H}_2\text{O}}^{\text{sat}}(T)$  where the saturation pressure is given by Antoine’s model as in Problem. 3.

Assuming ideal gas and constant temperature, and known influent flow rate  $\dot{V}_c^i$  of atmospheric air at pressure  $p_{\text{atm}}$  (containing oxygen and nitrogen; mole fraction  $x_{\text{O}_2}^{\text{air}} \approx 0.2$ ) controlled by the fan/compressor, the oxygen species balance leads to the following model for oxygen pressure:

|    | <b>Problem 12:</b> Dynamic model for oxygen pressure under given conditions — which model is <i>correct</i> ?                                                                                                     | Choice |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\frac{dp_{\text{O}_2}}{dt} = \frac{x_{\text{O}_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{\text{O}_2}}{V_c} \dot{V}_c^e - \frac{RT}{V_c} \dot{n}_{\text{O}_2}^g$                                |        |
| B: | $\frac{dp_{\text{O}_2}}{dt} = \frac{p_{\text{O}_2}}{V_c} \dot{V}_c^i - \frac{x_{\text{O}_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^e - \frac{RT}{V_c} \dot{n}_{\text{O}_2}^g$                                |        |
| C: | $\frac{dp_{\text{O}_2}}{dt} = \frac{x_{\text{O}_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{\text{O}_2}}{V_c} \dot{V}_c^e + \frac{RT}{V_c} \dot{n}_{\text{O}_2}^g$                                |        |
| D: | $\frac{dp_{\text{O}_2}}{dt} = \frac{x_{\text{O}_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{\text{O}_2}}{V_c} \dot{V}_c^e + \frac{RT}{V_c} (\dot{n}_{\text{H}_2\text{O}} - \dot{n}_{\text{N}_2})$ |        |

▲

### Solution 12. Cathode oxygen balance.

For Problem 12: The oxygen species balance is

$$\frac{dn_{\text{O}_2}}{dt} = \dot{n}_{\text{O}_2}^i - \dot{n}_{\text{O}_2}^e + \dot{n}_{\text{O}_2}^g.$$

With

$$\begin{aligned} p_{\text{O}_2} V_c &= n_{\text{O}_2} RT \\ p_{\text{O}_2}^{\text{air}} \dot{V}_c^i &= \dot{n}_{\text{O}_2}^i RT \\ p_{\text{O}_2}^{\text{air}} &= x_{\text{O}_2}^{\text{air}} p_{\text{atm}} \\ p_{\text{O}_2} \dot{V}_c^e &= \dot{n}_{\text{O}_2}^e RT \end{aligned}$$

we find

$$\begin{aligned}
\frac{dp_{O_2}}{dt} &= \frac{RT}{V_c} \left( \frac{p_{O_2}^{\text{air}} \dot{V}_c^i}{RT} - \frac{p_{O_2} \dot{V}_c^e}{RT} + \dot{n}_{O_2}^g \right) \\
&\Downarrow \\
\frac{dp_{O_2}}{dt} &= \frac{RT}{V_c} \left( \frac{x_{O_2}^{\text{air}} p_{\text{atm}} \dot{V}_c^i}{RT} - \frac{p_{O_2} \dot{V}_c^e}{RT} + \dot{n}_{O_2}^g \right) \\
&\Downarrow \\
\frac{dp_{O_2}}{dt} &= \frac{x_{O_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{O_2}}{V_c} \dot{V}_c^e + \frac{RT}{V_c} \dot{n}_{O_2}^g.
\end{aligned}$$

|    |                                                                                                                                                                             |        |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|    | <b>Solution 12:</b> Dynamic model for oxygen pressure under given conditions — which model is <i>correct</i> ?                                                              | Choice |
| A: | $\frac{dp_{O_2}}{dt} = \frac{x_{O_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{O_2}}{V_c} \dot{V}_c^e - \frac{RT}{V_c} \dot{n}_{O_2}^g$                      |        |
| B: | $\frac{dp_{O_2}}{dt} = \frac{p_{O_2}}{V_c} \dot{V}_c^i - \frac{x_{O_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^e - \frac{RT}{V_c} \dot{n}_{O_2}^g$                      |        |
| C: | $\frac{dp_{O_2}}{dt} = \frac{x_{O_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{O_2}}{V_c} \dot{V}_c^e + \frac{RT}{V_c} \dot{n}_{O_2}^g$                      | ×      |
| D: | $\frac{dp_{O_2}}{dt} = \frac{x_{O_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{O_2}}{V_c} \dot{V}_c^e + \frac{RT}{V_c} (\dot{n}_{H_2O}^g - \dot{n}_{N_2}^g)$ |        |

▲

**Problem 13.** Cathode nitrogen balance.

With the assumptions in Problem 12, the nitrogen species balance leads to the following model for nitrogen pressure:

|    |                                                                                                                                                                             |        |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|    | <b>Problem 13:</b> Dynamic model for nitrogen pressure under given conditions — which model is simplest and <i>correct</i> ?                                                | Choice |
| A: | $\frac{dp_{N_2}}{dt} = \frac{x_{O_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e$                                                       |        |
| B: | $\frac{dp_{N_2}}{dt} = \frac{x_{O_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e - \frac{RT}{V_c} (\dot{n}_{H_2O}^g - \dot{n}_{O_2}^g)$ |        |
| C: | $\frac{dp_{N_2}}{dt} = \frac{(1-x_{O_2}^{\text{air}}) p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e$                                                   |        |
| D: | $\frac{dp_{N_2}}{dt} = \frac{(1-x_{O_2}^{\text{air}}) p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e + \frac{RT}{V_c} \dot{n}_{N_2}^g$                  |        |

▲

**Solution 13.** Cathode nitrogen balance.

For Problem 13 — see Solution 12 for the oxygen balance, where  $x_{N_2}^{\text{air}} = 1 - x_{O_2}^{\text{air}}$  and  $\dot{n}_{N_2}^g \equiv 0$ .

|    |                                                                                                                                                                             |        |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|    | <b>Solution 13:</b> Dynamic model for nitrogen pressure under given conditions — which model is simplest and <i>correct</i> ?                                               | Choice |
| A: | $\frac{dp_{N_2}}{dt} = \frac{x_{O_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e$                                                       |        |
| B: | $\frac{dp_{N_2}}{dt} = \frac{x_{O_2}^{\text{air}} p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e - \frac{RT}{V_c} (\dot{n}_{H_2O}^g - \dot{n}_{O_2}^g)$ |        |
| C: | $\frac{dp_{N_2}}{dt} = \frac{(1-x_{O_2}^{\text{air}}) p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e$                                                   | ×      |
| D: | $\frac{dp_{N_2}}{dt} = \frac{(1-x_{O_2}^{\text{air}}) p_{\text{atm}}}{V_c} \dot{V}_c^i - \frac{p_{N_2}}{V_c} \dot{V}_c^e + \frac{RT}{V_c} \dot{n}_{N_2}^g$                  |        |

▲

**Problem 14.** Cathode effluent flow rate.

We will assume that  $\dot{V}_c^i$  is a known function of time. From Bernoulli's steady state energy balance, a typical valve model for the *effluent* flow rate would be

$$\dot{V}_c^e = K_c \cdot \sqrt{\frac{\Delta p_c^e}{p^o}}. \quad (23)$$

Consider the term  $\dot{V}_c^e$  in Problems 12–13. Modify the model in Eq. 23 so that (i) the flow unidirectional (i.e., only can flow out), and (ii) give expressions for  $\Delta p_c^e$ .

|    | <b>Problem 14:</b> Modified cathode effluent flow rate $\dot{V}_c^a$ — which set of expressions is <i>correct</i> ?                                                       | Choice |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\dot{V}_c^e = K_c \cdot \sqrt{\max(\Delta p_c^e, 0) / p^o}$ ; $\Delta p_c^e = p_c - p_{\text{atm}}$ ; $p_c = p_{\text{O}_2} + p_{\text{N}_2} + p_{\text{H}_2\text{O}}^c$ |        |
| B: | $\dot{V}_c^e = K_c \cdot \sqrt{\min(\Delta p_c^e, 0) / p^o}$ ; $\Delta p_c^e = p_c - p_{\text{atm}}$ ; $p_c = p_{\text{O}_2} + p_{\text{N}_2} + p_{\text{H}_2\text{O}}^c$ |        |
| C: | $\dot{V}_c^e = K_c \cdot \sqrt{\max(\Delta p_c^e, 0) / p^o}$ ; $\Delta p_c^e = p_c - p_{\text{atm}}$ ; $p_c = p_{\text{O}_2} + p_{\text{N}_2}$                            |        |
| D: | $\dot{V}_c^e = K_c \cdot \sqrt{\min(\Delta p_c^e, 0) / p^o}$ ; $\Delta p_c^e = p_c - p_{\text{atm}}$ ; $p_c = p_{\text{O}_2} + p_{\text{N}_2}$                            |        |

▲

**Solution 14.** Cathode effluent flow rate.

For Problem 14: this is relatively straightforward...

|    | <b>Solution 14:</b> Modified cathode effluent flow rate $\dot{V}_c^a$ — which set of expressions is <i>correct</i> ?                                                      | Choice |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\dot{V}_c^e = K_c \cdot \sqrt{\max(\Delta p_c^e, 0) / p^o}$ ; $\Delta p_c^e = p_c - p_{\text{atm}}$ ; $p_c = p_{\text{O}_2} + p_{\text{N}_2} + p_{\text{H}_2\text{O}}^c$ | ×      |
| B: | $\dot{V}_c^e = K_c \cdot \sqrt{\min(\Delta p_c^e, 0) / p^o}$ ; $\Delta p_c^e = p_c - p_{\text{atm}}$ ; $p_c = p_{\text{O}_2} + p_{\text{N}_2} + p_{\text{H}_2\text{O}}^c$ |        |
| C: | $\dot{V}_c^e = K_c \cdot \sqrt{\max(\Delta p_c^e, 0) / p^o}$ ; $\Delta p_c^e = p_c - p_{\text{atm}}$ ; $p_c = p_{\text{O}_2} + p_{\text{N}_2}$                            |        |
| D: | $\dot{V}_c^e = K_c \cdot \sqrt{\min(\Delta p_c^e, 0) / p^o}$ ; $\Delta p_c^e = p_c - p_{\text{atm}}$ ; $p_c = p_{\text{O}_2} + p_{\text{N}_2}$                            |        |

▲

**Problem 15.** Cathode generation.

Consider the term  $\dot{n}_{\text{O}_2}^g$  in Problem 12. With  $\nu = (\nu_{\text{H}_2}, \nu_{\text{O}_2}, \nu_{\text{H}_2\text{O}})$ , what is the correct expression for  $\dot{n}_{\text{O}_2}^g$  for a single fuel cell?

|    | <b>Problem 15:</b> Which of the following expressions for oxygen generation $\dot{n}_{\text{O}_2}^g$ is <i>correct</i> ? | Choice |
|----|--------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\dot{n}_{\text{O}_2}^g = \nu_{\text{O}_2} \cdot i / (F \nu_{e^-})$                                                      |        |
| B: | $\dot{n}_{\text{O}_2}^g = -\nu_{\text{O}_2} \cdot i / (F \nu_{e^-})$                                                     |        |
| C: | $\dot{n}_{\text{O}_2}^g = \nu_{\text{O}_2} \cdot i / F$                                                                  |        |
| D: | $\dot{n}_{\text{O}_2}^g = -\nu_{\text{O}_2} \cdot i / F$                                                                 |        |

▲

**Solution 15.** Cathode generation.

For Problem 15: oxygen consumption is related to hydrogen consumption as

$$\dot{n}_{\text{O}_2}^g = \frac{\nu_{\text{O}_2}}{\nu_{\text{H}_2}} \dot{n}_{\text{H}_2}^g = \frac{\nu_{\text{O}_2}}{\nu_{\text{H}_2}} \cdot \frac{\nu_{\text{H}_2}}{F \nu_{e^-}} i = \frac{\nu_{\text{O}_2}}{F \nu_{e^-}} i.$$

|    |                                                                                                                           |        |
|----|---------------------------------------------------------------------------------------------------------------------------|--------|
|    | <b>Solution 15:</b> Which of the following expressions for oxygen generation $\dot{n}_{\text{O}_2}^g$ is <i>correct</i> ? | Choice |
| A: | $\dot{n}_{\text{O}_2}^g = \nu_{\text{O}_2} \cdot i / (F \nu_{e-})$                                                        | ×      |
| B: | $\dot{n}_{\text{O}_2}^g = -\nu_{\text{O}_2} \cdot i / (F \nu_{e-})$                                                       |        |
| C: | $\dot{n}_{\text{O}_2}^g = \nu_{\text{O}_2} \cdot i / F$                                                                   |        |
| D: | $\dot{n}_{\text{O}_2}^g = -\nu_{\text{O}_2} \cdot i / F$                                                                  |        |

▲

**Problem 16.** Cathode diffusion/current limitation.

There is a possible gas pressure drop from the cathode volume to the surface of the catalyst/membrane due to diffusion resistance through the Gas Diffusion Layer (Fig. 6), etc. With water vapor, oxygen, and nitrogen in the cathode volume, a possible expression for the resulting pressure drop in oxygen pressure is

$$\frac{p_{\text{O}_2}^{\text{cs}}}{p_{\text{O}_2}^{\text{c}}} = \frac{p_{\text{c}} - p_{\text{H}_2\text{O}}^{\text{c}} - p_{\text{N}_2}^{\text{c}} \exp(f_{\text{c}}(T) \cdot i'')}{p_{\text{c}} - p_{\text{H}_2\text{O}}^{\text{c}} - p_{\text{N}_2}^{\text{c}}} \quad (24)$$

where superscripts c and cs refer to *bulk* cathode volume and *surface* (i.e., at the catalyst/membrane surface),  $p_{\text{H}_2\text{O}}^{\text{c}} = p_{\text{H}_2\text{O}}^{\text{sat}}(T)$ ,  $f_{\text{c}}(T) > 0$  is a known function of temperature (involving diffusion coefficients, etc.), while  $i''$  is the current density in the membrane.

In the literature, it is common to write this reduction in partial pressure as

$$\frac{p_{\text{O}_2}^{\text{cs}}}{p_{\text{O}_2}^{\text{c}}} \approx 1 - \frac{i''}{i_{\ell}''}$$

where  $i_{\ell}''$  is a *limiting* current density.

Find an expression for  $i_{\ell}''$  based on (linearization of) Eq. 24.

|    |                                                                                                            |        |
|----|------------------------------------------------------------------------------------------------------------|--------|
|    | <b>Problem 16:</b> Which of the following expressions for $i_{\ell}''$ is <i>consistent</i> with Eq. 24?   | Choice |
| A: | $i_{\ell}'' = \frac{p_{\text{N}_2}^{\text{c}}}{p_{\text{O}_2}^{\text{c}}} \cdot f_{\text{c}}(T)$           |        |
| B: | $i_{\ell}'' = \frac{p_{\text{N}_2}^{\text{c}}}{p_{\text{O}_2}^{\text{c}}} \cdot \frac{1}{f_{\text{c}}(T)}$ |        |
| C: | $i_{\ell}'' = \frac{p_{\text{O}_2}^{\text{c}}}{p_{\text{N}_2}^{\text{c}}} \cdot f_{\text{c}}(T)$           |        |
| D: | $i_{\ell}'' = \frac{p_{\text{O}_2}^{\text{c}}}{p_{\text{N}_2}^{\text{c}}} \cdot \frac{1}{f_{\text{c}}(T)}$ |        |

▲

**Solution 16.** Cathode diffusion/current limitation.

For Problem 16: this is relatively straightforward... in Eq. 24,

$$\exp(f_{\text{c}}(T) \cdot i'') \approx 1 + f_{\text{c}}(T) i'',$$

leading to the following approximation for Eq. 24:

$$\begin{aligned}
\frac{p_{\text{O}_2}^{\text{cs}}}{p_{\text{O}_2}^{\text{c}}} &\approx \frac{p_{\text{c}} - p_{\text{H}_2\text{O}}^{\text{c}} - p_{\text{N}_2}^{\text{c}} (1 + f_{\text{c}}(T) i'')}{p_{\text{c}} - p_{\text{H}_2\text{O}}^{\text{c}} - p_{\text{N}_2}^{\text{c}}} \\
&\Downarrow \\
\frac{p_{\text{O}_2}^{\text{cs}}}{p_{\text{O}_2}^{\text{c}}} &\approx \frac{p_{\text{c}} - p_{\text{H}_2\text{O}}^{\text{c}} - p_{\text{N}_2}^{\text{c}} - p_{\text{N}_2}^{\text{c}} f_{\text{c}}(T) i''}{\underbrace{p_{\text{c}} - p_{\text{H}_2\text{O}}^{\text{c}} - p_{\text{N}_2}^{\text{c}}}_{=p_{\text{O}_2}^{\text{c}}}} \\
&\Downarrow \\
\frac{p_{\text{O}_2}^{\text{cs}}}{p_{\text{O}_2}^{\text{c}}} &\approx \frac{p_{\text{c}} - p_{\text{H}_2\text{O}}^{\text{c}} - p_{\text{N}_2}^{\text{c}} - p_{\text{N}_2}^{\text{c}} f_{\text{c}}(T) i''}{p_{\text{O}_2}^{\text{c}}} = 1 - \frac{p_{\text{N}_2}^{\text{c}}}{p_{\text{O}_2}^{\text{c}}} f_{\text{c}}(T) i''.
\end{aligned}$$

It follows that

$$i''_{\ell} = \frac{p_{\text{O}_2}^{\text{c}}}{p_{\text{N}_2}^{\text{c}}} \cdot \frac{1}{f_{\text{c}}(T)}.$$

|    | <b>Solution 16:</b> Which of the following expressions for $i''_{\ell}$ is <i>consistent</i> with Eq. 24?  | Choice |
|----|------------------------------------------------------------------------------------------------------------|--------|
| A: | $i''_{\ell} = \frac{p_{\text{N}_2}^{\text{c}}}{p_{\text{O}_2}^{\text{c}}} \cdot f_{\text{c}}(T)$           |        |
| B: | $i''_{\ell} = \frac{p_{\text{N}_2}^{\text{c}}}{p_{\text{O}_2}^{\text{c}}} \cdot \frac{1}{f_{\text{c}}(T)}$ |        |
| C: | $i''_{\ell} = \frac{p_{\text{O}_2}^{\text{c}}}{p_{\text{N}_2}^{\text{c}}} \cdot f_{\text{c}}(T)$           |        |
| D: | $i''_{\ell} = \frac{p_{\text{O}_2}^{\text{c}}}{p_{\text{N}_2}^{\text{c}}} \cdot \frac{1}{f_{\text{c}}(T)}$ | ×      |

▲

**Problem 17.** Maximal Fuel Cell efficiency.

Fuel Cell efficiency  $\eta$  is typically found as the produced electric power divided by the enthalpy of reaction,

$$\eta = \frac{u \cdot i}{\Delta_{\text{r}} \tilde{H} \cdot \dot{n}_{\text{H}_2\text{O}}^{\text{g}}}.$$

Here, current  $i$  is given as

$$i = -F \cdot \dot{n}_{\text{e}^-}$$

where  $\dot{n}_{\text{e}^-}$  is the molar flow rate of electrons, and the generation of product  $\dot{n}_{\text{H}_2\text{O}}^{\text{g}}$  is related to  $\dot{n}_{\text{e}^-}$  as

$$\dot{n}_{\text{H}_2\text{O}}^{\text{g}} = \frac{\dot{n}_{\text{e}^-}}{\nu_{\text{e}^-}}.$$

In the limit  $i \rightarrow 0^+$ ,  $u = u_{\text{n}}$  where Nernst voltage  $u_{\text{n}}$  is as given in Problem 2. Denote the resulting maximal efficiency by  $\eta_0$  [zero current],

$$\eta_0 = \frac{u_{\text{n}} \cdot i}{\Delta_{\text{r}} \tilde{H} \cdot \dot{n}_{\text{H}_2\text{O}}^{\text{g}}}.$$

For simplicity, we consider a reference temperature of 25 °C, which gives

$$\Delta_{\text{r}} \tilde{G} = \Delta_{\text{r}} \tilde{G}^{\circ} = 1 \cdot \Delta_{\text{f}} \tilde{G}_{\text{H}_2\text{O}} = -228.572 \text{ kJ/mol}$$

while

$$\Delta_r \tilde{H}^\circ = -241.785 \text{ kJ/mol.}$$

Based upon the above considerations, what is the theoretical maximal efficiency of a fuel cell at standard state (i.e., pressures  $p_j = p^\circ$ ) and with (extremely) low current throughput  $i$ , when we assume that both reactants and product are in gas phase?

|    |                                                                                                                                            |        |
|----|--------------------------------------------------------------------------------------------------------------------------------------------|--------|
|    | <b>Problem 17:</b> Approximate fuel cell efficiency under the above considerations — which maximal efficiency $\eta_0$ is <i>correct</i> ? | Choice |
| A: | $\eta_0 \approx 0.4$                                                                                                                       |        |
| B: | $\eta_0 \approx 0.95$                                                                                                                      |        |
| C: | $\eta_0 \approx 0.83$                                                                                                                      |        |
| D: | $\eta_0 \approx 0.28$                                                                                                                      |        |

▲

**Solution 17.** Maximal Fuel Cell efficiency.

For Problem 17: Fuel Cell efficiency  $\eta$  is typically found as the produced electric power divided by the enthalpy of reaction,

$$\eta = \frac{u \cdot i}{\Delta_r \tilde{H} \cdot \dot{n}_{\text{H}_2\text{O}}^g}.$$

Here, for simplicity assume that  $u = u_n$  where Nernst voltage  $u_n$  is as given in Problem 2, while

$$i = -F \cdot \dot{n}_{e^-}$$

and

$$\dot{n}_{\text{H}_2\text{O}}^g = \frac{\dot{n}_{e^-}}{\nu_{e^-}}.$$

This gives

$$\eta = \frac{u \cdot i}{\Delta_r \tilde{H} \cdot \dot{n}_{\text{H}_2\text{O}}^g} = \frac{-u \cdot F \cdot \dot{n}_{e^-}}{\Delta_r \tilde{H} \cdot \frac{\dot{n}_{e^-}}{\nu_{e^-}}} = \frac{u F \nu_{e^-}}{-\Delta_r \tilde{H}}.$$

From the expression for Nernst voltage at standard state,  $u = u_o$  (Problem ), we find

$$u_o = \frac{-\Delta_r \tilde{G}^\circ}{F \nu_{e^-}},$$

and therefore

$$\eta = \frac{\frac{-\Delta_r \tilde{G}^\circ}{F \nu_{e^-}} F \nu_{e^-}}{-\Delta_r \tilde{H}^\circ} = \frac{\Delta_r \tilde{G}^\circ}{\Delta_r \tilde{H}^\circ}.$$

For simplicity, at reference temperature  $T$  equivalent to 25 °C, we then have

$$\Delta_r \tilde{G} = \Delta_r \tilde{G}^\circ = 1 \cdot \Delta_r \tilde{G}_{\text{H}_2\text{O}} = -228.572 \text{ kJ/mol}$$

while

$$\Delta_r \tilde{H} = \Delta_r \tilde{H}^\circ = \Delta_r \tilde{G}^\circ + T^\circ \cdot \Delta_r \tilde{S}^\circ.$$

Here,

$$\Delta_r \tilde{S}^\circ = \nu \cdot \begin{pmatrix} \Delta_f \tilde{S}_{\text{H}_2}^\circ \\ \Delta_f \tilde{S}_{\text{O}_2}^\circ \\ \Delta_f \tilde{S}_{\text{H}_2\text{O}}^\circ \end{pmatrix} = \begin{pmatrix} -1 & -\frac{1}{2} & 1 \end{pmatrix} \begin{pmatrix} 130.5 \\ 205.03 \\ 188.7 \end{pmatrix}$$

$$\Downarrow$$

$$\Delta_r \tilde{S}^\circ \approx -44.315 \text{ J/mol K.}$$

It follows that

$$\Delta_r \tilde{H}^\circ = -228.572 \cdot 10^3 + (25 + 273.15) \cdot (-44.315) = -241.785 \text{ kJ/mol.}$$

The efficiency is thus

$$\eta = \frac{-228.572}{-241.785} \approx 0.95.$$

This efficiency assumes that the current flow is very low, and that the reaction takes place in gas phase, and that the water product is in gas phase. In practice, water will come out of the Fuel Cell as liquid, which means that the condensation enthalpy will be lost, thus reducing the efficiency to around  $\eta = 0.83$ .

|    | <b>Solution 17:</b> Approximate fuel cell efficiency under the above considerations — which maximal efficiency $\eta_0$ is <i>correct</i> ? | Choice |
|----|---------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\eta_0 \approx 0.4$                                                                                                                        |        |
| B: | $\eta_0 \approx 0.95$                                                                                                                       | ×      |
| C: | $\eta_0 \approx 0.83$                                                                                                                       |        |
| D: | $\eta_0 \approx 0.28$                                                                                                                       |        |

**Problem 18.** Practical Fuel Cell efficiency.

Figure 7 shows how the external voltage  $u$  in the efficiency expression of Problem 17 varies with current density  $i''$  (and thus with  $i$ ).

Based on the above, how much is the practical efficiency  $\eta_{i''}$  of a fuel cell *reduced* compared to the maximal efficiency, say at  $i'' = 0.5 \text{ A/cm}^2$ ?

|    | <b>Problem 18:</b> Approximate fuel cell efficiency under the above considerations — which efficiency $\eta_{0.5}$ is <i>correct</i> ? | Choice |
|----|----------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | $\eta_{0.5} \approx \eta_0$                                                                                                            |        |
| B: | $\eta_{0.5} \approx 0.49 \cdot \eta_0$                                                                                                 |        |
| C: | $\eta_{0.5} \approx 0.61 \cdot \eta_0$                                                                                                 |        |
| D: | $\eta_{0.5} \approx 0.83 \cdot \eta_0$                                                                                                 |        |

**Solution 18.** Practical Fuel Cell efficiency.

For Problem 18: The Nernst equilibrium voltage has a value of approximately 1.17 V. At  $i'' = 0.5 \text{ A/cm}^2$ , from Fig. 7,  $u \approx 0.57 \text{ V}$ . This gives

$$\eta_{0.5} \approx \frac{0.57}{1.17} \approx 0.49.$$

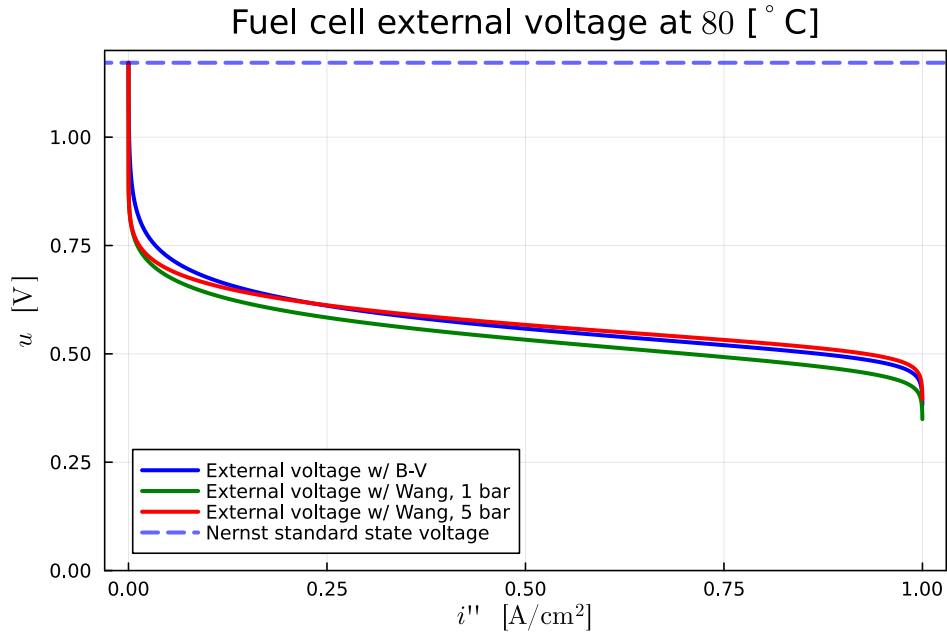


Figure 7: Typical external voltage of fuel cell: (i) using Butler-Volmer activation voltage, (ii)-(iii) using a more realistic activation voltage due to Wang, at 1 and 5 bar, respectively, (iv) with zero current/Nernst voltage (blue, dashed) .

|    |                                                                                                                                         |        |
|----|-----------------------------------------------------------------------------------------------------------------------------------------|--------|
|    | <b>Solution 18:</b> Approximate fuel cell efficiency under the above considerations — which efficiency $\eta_{0.5}$ is <i>correct</i> ? | Choice |
| A: | $\eta_{0.5} \approx \eta_0$                                                                                                             |        |
| B: | $\eta_{0.5} \approx 0.49 \cdot \eta_0$                                                                                                  | ×      |
| C: | $\eta_{0.5} \approx 0.61 \cdot \eta_0$                                                                                                  |        |
| D: | $\eta_{0.5} \approx 0.83 \cdot \eta_0$                                                                                                  |        |

▲

**Problem 19.** Consequence of efficiency loss.

In Problems 17–18, we have seen that a fuel cell does not have 100% conversion of reactants into electric power.

What is the consequence of imperfect efficiency  $\eta$ ?

|    | <b>Problem 19:</b> Which of the following statements is <i>correct</i> ?                                                                                                                                                                                                               | Choice |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | Both with $i'' \rightarrow 0^+$ ( $\eta_0$ ) and for finite $i''$ ( $\eta_{i''}$ ) will the efficiency loss lead to heating of the fuel cell. It is necessary to actively control the cooling of the fuel cell to (i) avoid overheating, and (ii) manage the humidity of the membrane. |        |
| B: | The efficiency loss with $i'' \rightarrow 0^+$ ( $\eta_0$ ) is an <i>inherent</i> loss, that has no effect on the temperature of the system; there can be no loss when there is no current.                                                                                            |        |
| C: | The efficiency loss for finite $i'' > 0$ ( $\eta_{i''}$ ) leads to heating of the fuel cell, and some cooling is necessary. However, this does not influence the humidity of the system.                                                                                               |        |
| D: | The efficiency loss for finite $i'' > 0$ ( $\eta_{i''}$ ) leads to heating of the fuel cell, but the heating is automatically compensated for by the humidity/water balance.                                                                                                           |        |

▲

**Solution 19.** Consequence of efficiency loss.

For Problem 19: because  $\eta_0 < 1$ , heat is produced even when  $i'' \equiv 0$ . As  $i''$  increases, the produced heat increases rapidly because the curve  $u(i'')$  in Fig. 7 drops steeply. Overheating makes the temperature unstable, and it is difficult to get a constant voltage. Furthermore, heating will make liquid water evaporate, and this will dry out the membrane and effectively make the membrane resistance increase dramatically until the fuel cell stops working. In summary, it is *necessary* to manage both the temperature and humidity of the membrane independently.

|    | <b>Solution 19:</b> Which of the following statements is <i>correct</i> ?                                                                                                                                                                                                              | Choice |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | Both with $i'' \rightarrow 0^+$ ( $\eta_0$ ) and for finite $i''$ ( $\eta_{i''}$ ) will the efficiency loss lead to heating of the fuel cell. It is necessary to actively control the cooling of the fuel cell to (i) avoid overheating, and (ii) manage the humidity of the membrane. | ×      |
| B: | The efficiency loss with $i'' \rightarrow 0^+$ ( $\eta_0$ ) is an <i>inherent</i> loss, that has no effect on the temperature of the system; there can be no loss when there is no current.                                                                                            |        |
| C: | The efficiency loss for finite $i'' > 0$ ( $\eta_{i''}$ ) leads to heating of the fuel cell, and some cooling is necessary. However, this does not influence the humidity of the system.                                                                                               |        |
| D: | The efficiency loss for finite $i'' > 0$ ( $\eta_{i''}$ ) leads to heating of the fuel cell, but the heating is automatically compensated for by the humidity/water balance.                                                                                                           |        |

▲

**Problem 20.** Choice of operating conditions.

As seen from the expression for pressure correction to the standard state equilibrium voltage, increasing the pressure (if possible) should give a higher equilibrium voltage, and hence more power out of the fuel cell.

The Butler-Volmer expression for activation voltage used in the group project was pressure independent; however, a more accurate (and complex) expression for the cathode

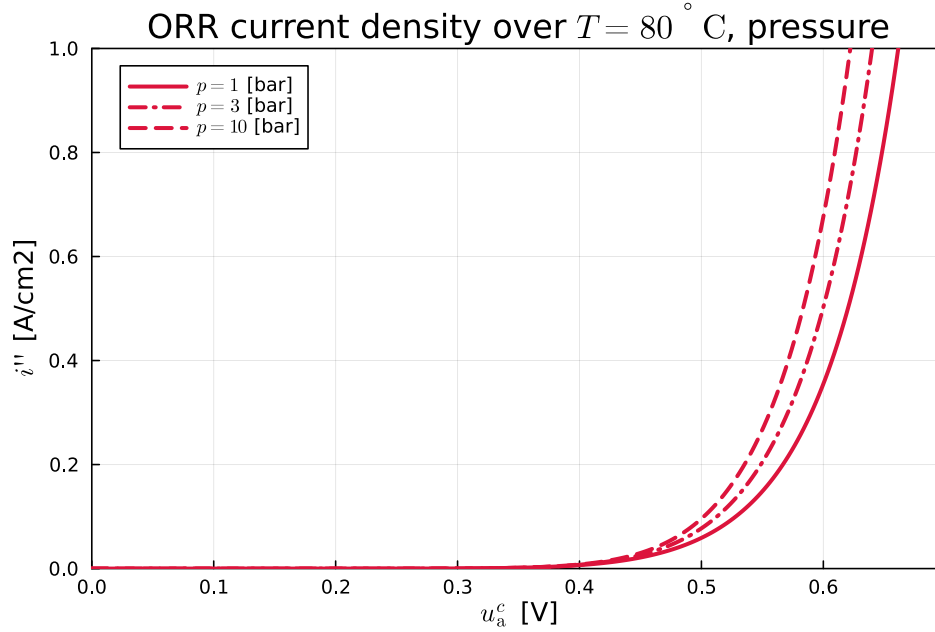


Figure 8: Pressure dependence of activation voltage loss relation.

activation, Fig. 8, shows that the voltage loss  $u_a$  for a *given* current density  $i''$  *decreases* with pressure.

In summary: the equilibrium voltage *increases* with higher electrode pressures, and the activation *losses* are *reduced* when current starts to flow. This means that there is a *double advantage* in operating at higher pressure.

Which of the following arguments for choice of operating pressures has potential safety issues?

|    | <b>Problem 20:</b> Choice of operating pressures for fuel cell — which argument has <i>safety issues</i> ?                                                                                                                                                                              | Choice |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | Operating at a high pressure requires power to compress the reactants (hydrogen, oxygen). The chosen pressure conditions must balance the increased electric power produced vs. the increased power needed to compress the reactants.                                                   |        |
| B: | Increased pressures in the electrode volumes must be balanced against the possibility of gas permeation through the membrane; higher pressure may require thicker membrane.                                                                                                             |        |
| C: | Operating at a high pressure may require more expensive equipment (thicker membrane to avoid gas permeation, also leading to higher resistive voltage loss $u_r$ ), more solid build, etc., and the increased design/purchase cost must be balanced against increased power production. |        |
| D: | The best way to increase the power is to increase the Nernst voltage by feeding oxygen to the cathode instead of air. This way, the partial pressure of oxygen can be made much higher while still having the same electrode pressures, $p_a = p_c$ .                                   |        |

▲

**Solution 20.** Choice of operating conditions.

For Problem 20: using pure oxygen as cathode feed is theoretically very beneficial. With oxygen feed, the partial pressure of oxygen will increase approximately 5 times compared to when using air. This will both give increased equilibrium voltage, and reduced activation loss — a double benefit. In addition, if hydrogen is produced via electrolysis of water, oxygen which is produced in the same process, and could be stored and later be used in a fuel cell: this would also increase the efficiency of electrolysis (otherwise, oxygen is released to air and essentially wasted).

However, because a possible leakage of pure oxygen (or permeation through the membrane) in an environment which already is challenged by presence of pure hydrogen — makes the use of pure oxygen feed a safety issue: in case of leakage, this will essentially give a close to pure mixture of hydrogen and oxygen, which is extremely explosive.

|    | <b>Solution 20:</b> Choice of operating pressures for fuel cell — which argument has <i>safety issues</i> ?                                                                                                                                                                             | Choice |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| A: | Operating at a high pressure requires power to compress the reactants (hydrogen, oxygen). The chosen pressure conditions must balance the increased electric power produced vs. the increased power needed to compress the reactants.                                                   |        |
| B: | Increased pressures in the electrode volumes must be balanced against the possibility of gas permeation through the membrane; higher pressure may require thicker membrane.                                                                                                             |        |
| C: | Operating at a high pressure may require more expensive equipment (thicker membrane to avoid gas permeation, also leading to higher resistive voltage loss $u_r$ ), more solid build, etc., and the increased design/purchase cost must be balanced against increased power production. |        |
| D: | The best way to increase the power is to increase the Nernst voltage by feeding oxygen to the cathode instead of air. This way, the partial pressure of oxygen can be made much higher while still having the same electrode pressures, $p_a = p_c$ .                                   | ×      |

▲

— END OF EXAM —